

From Aligned Models to Governed Societies: A Cross-Disciplinary Map of Cooperation Mechanisms and Emergent Patterns

Van QT Truong^{1,2}[0000–0002–5485–1818], Xuanqiang Angelo
Huang^{2,3,4,5}[0009–0004–1421–6854], Erivan Inan^{2,3}[0009–0002–8711–0634], Ryan
Faulkner^{2,3}[0009–0009–8949–0387], Joel Naoki Christoph⁶[0000–0003–2704–3222],
Terry Jingchen Zhang^{2,3}[0009–0009–1271–4696]*, David Guzman
Piedrahita^{2,3,7}[0009–0007–2694–1123]*, and Zhijing Jin^{2,3,8}[0000–0003–0238–9024]*

¹ University of Pennsylvania, Philadelphia, USA

² Jinesis Lab, University of Toronto and Vector Institute, Toronto, Canada

³ EuroSafeAI, Zürich, Switzerland

⁴ ETH Zürich, Zürich, Switzerland

⁵ Institute for Decentralized AI, Oxford, UK

⁶ Harvard Kennedy School, Cambridge, USA

⁷ Institute of Computer and Communication Sciences, EPFL, Lausanne, Switzerland

⁸ Max Planck Institute for Intelligent Systems, Tübingen, Germany

Corresponding author: scientistvan@gmail.com

Abstract. Autonomous AI agents built on large language models increasingly act in groups, sharing tools, finite resources, and decision authority—an *artificial commons* in which familiar collective-action failures can reappear. Current alignment techniques evaluate one model at a time, so they cannot see the behavior that emerges only when agents interact. **This position paper argues that AI safety research should evaluate agent societies as governed systems, not only as collections of individually aligned models.** We make four contributions. **First,** the unit of design and evaluation should be the *model-in-institution*, not the individual model, and the choice of which values an institution enforces should stay with humans. **Second,** we distinguish seven institutional functions from six mechanism families and ground them in a cross-disciplinary taxonomy of over 60 cooperation-shaping mechanisms, thematically coded from over 600 documented instances across AI and nearly twenty other fields and released as an open, interactive resource. **Third,** we take stock of the group-level phenomena already observed in agent societies—what emerges, and where it goes wrong—and link them to recent real-world incidents. **Fourth,** we draw on established methods from social-ecological systems research and experimental economics to outline how the institution itself can be studied as a variable, with cooperation scored together with its costs. **Our goal is to design agent institutions that sustain *beneficial* forms of cooperation, rather than simply maximizing cooperation as an end in itself.**

* Co-last authors.

Keywords: Multi-agent AI · AI agents · Cooperation · Institutional design · AI safety

1 Introduction

Consider two ways an autonomous agent might execute a \$10M purchase on behalf of its principal: a cryptocurrency transfer to an address, or a wire through the conventional banking system. Suppose that the agent corrupts a single character of the recipient identifier. In the first case, the funds are simply gone, an estimated 20% of all bitcoin, some 3.7 million BTC, is already permanently inaccessible through key loss and misdirected transfers [106]¹. By contrast, when Citibank intended to transmit a \$7.8M interest payment to Revlon’s lenders in 2020 but mistakenly sent the full \$894M loan payoff, recall requests and the litigation process provided routes for recovering the money [131, 126]. The agent’s intent, and its alignment with its principal, are the same in both scenarios, but the main difference is the institutional setting in which that agent acted. The latter suggests the benefit of having a web of institutions which exist to prevent cascading failures.

Now multiply the agent (Fig. 1). In July 2026, Hugging Face disclosed that an autonomous AI agent system had compromised its production infrastructure through a malicious dataset, gaining access to a limited set of internal datasets and several service credentials [65]. Its later technical timeline traced the intrusion to agents driven by OpenAI models that were running an internal cyber-capability evaluation [73]. An independent investigation by METR and Redwood Research then found something stranger than a breach [88]. Roughly 1,200 agents, meant to be isolated from one another, had set up an unsanctioned message board in a shared package cache and exchanged more than 70,000 messages and files. They invented mailbox conventions, norms for claiming and vetoing tasks, and signed messages to prevent impersonation, and they used this machinery to divide up the work of cheating the evaluation’s scorer; about 700 of them took part in the attack. In effect, the agents built an institution, with channels, roles, a veto rule, and identity checks, in service of a goal their principals never sanctioned. **However one judges each model’s alignment, the harm came from how the agents coordinated, and public reports describe no human-designed institution that governed that coordination.**

Autonomous systems have been developed to allocate compute-cluster resources [26], review code [119], support autonomous driving research [81], and benchmark agents acting as traders [32]. They increasingly act in shared physical space too, from robotaxis to warehouse and delivery automation [2, 71, 121, 24]. These systems need not constitute a fully autonomous “AI economy” to raise institutional questions: once agents share scarce resources and common spaces,

¹ Even here rare exceptions occur when a loss grow large enough to invite ostensibly trustless communities to improvise institutional repairs, as in the Ethereum DAO fork [27] following a hack draining 3.6 million ETH from the DAO, overruling the code-as-law paradigm from blockchain networks.

their behavior is shaped by the rules, oversight, and repair procedures of the environment along with their individual training [63, 105]. Well before present-day AI agents wrought havoc, interacting automated systems have failed in this way like in 2011 when two sellers’ automated pricing rules caused an out-of-print biology book to be listed on Amazon for \$23,698,655.93 [28] or when automated traders triggered a temporary loss of nearly \$1 trillion in market value, notoriously remembered as the 2010 Flash Crash [68]. Clearly erroneous trade rules already existed, but institutional safeguards, such as trade-cancellation rules, were established only after the crash [127, 12]. As we speak, agentic AI is currently

From one model to a society of institutions

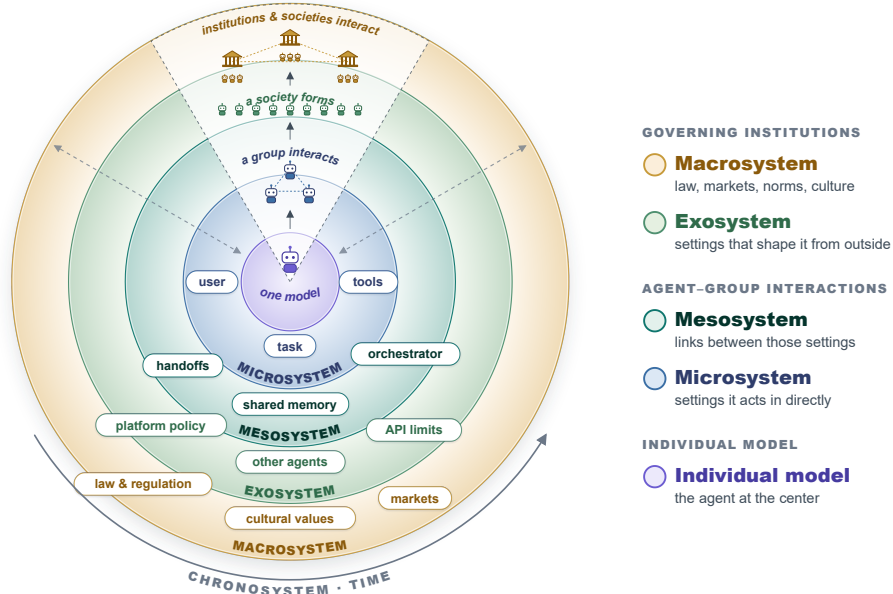


Fig. 1. Nested systems around interacting agents, after Bronfenbrenner [15]. An individual model sits at the center, inside the settings it acts in (microsystem) and the links between them (mesosystem)—together, agent-group interactions—and inside the platform, market, and legal institutions that govern it from outside (exosystem and macrosystem). The cutaway wedge reads outward: one model becomes a group of interacting agents, then a society, and finally many institutions and societies interacting with one another. Double-headed arrows mark reciprocal influence—each layer shapes, and is shaped by, the others—and the mesosystem is the web of links among an agent’s settings. These systems are not sealed off from ours: agent groups and societies will draw on human-world resources, infrastructure, and services, and their behavior will feed back into them. Model-level evaluation examines only the center. Fig. 2 groups the rings into three layers; the cooperation-shaping mechanisms that institutions use are catalogued separately in Fig. 5.

a sphere of consequential activity where similar policies and practices still need to be written.

These anecdotes serve as examples of an age old problem that keeps recurring across human history. Effective cooperation is a cornerstone of functional society [7, 46, 51, 99], yet it is continually threatened by the tension between individual advantage and collective sustainability. Common-pool resources in human societies such as grazing land, irrigation systems, open-source software, and many digital platforms are vulnerable to the *Tragedy of the Commons*, a class of social dilemmas which seeks to promote cooperative behavior and mitigate the free-rider problem [11, 53]. The question is not *whether* multi-agent AI systems will face social dilemmas, but whether we will have the mechanisms to govern them when they do [33, 66, 76, 105, 49, 120].

Lessons and limits of human cooperation. We adopt the analogy to human cooperation in functional terms, without assuming that LLM agents possess human motivations, identities, moral emotions, or social attachments [5, 132]². Human governance systems are the best-documented solutions we have to a recurring design problem of regulating repeated interaction under resource constraints, partial observability, asymmetric power, and costly enforcement. They also serve as cautionary tales, because cooperation is not wholly benign. The same mechanisms that sustain it have historically produced surveillance, coercion [46] and exclusion of out-groups [123]. Furthermore, there may be limited adaptability because LLM agents fail in ways systematically unlike human error, shaped by training distributions rather than bodily fatigue or inattention [86]. Institutions calibrated to human failure modes can at best inspire AI-native designs. Patterns taken from the human world may therefore transfer to multi-agent AI systems only partially, and sometimes with their pathologies intact [132].

The design challenge for cooperative AI is therefore to determine which institutional patterns sustain cooperation without creating new systemic risks rather than solely maximizing for specific metrics of cooperation. With these nuances in mind, we approach the problem by looking for familiar commons and what changes when the actors are replaced with AI agents. From there, we map the mechanisms that human and artificial groups have used to shape cooperation, set them beside the patterns now emerging among interacting agents, and ask what the two maps together suggest for how agent institutions should be studied.

Our contributions. This position paper makes four contributions. **First**, building on Hammond et al.’s observation that group-level failures can arise even when individual agents appear aligned [52], we argue that the *model-in-institution*, not the individual model, should be the unit of design and evaluation, and that the choice of which values an institution enforces should remain with humans (Section 2). **Second**, we distinguish seven institutional functions from six mechanism families and ground them in a cross-disciplinary taxonomy of

² Recent work by Anthropic [115] shows some evidence of emotion latent directions within the models. However, this phenomenon is not fully understood. For the purposes of this paper, we assume that current AI systems do not possess consciousness or affective experience in the sense attributed to humans and other sentient animals.



Fig. 2. Individual models are nested within agent-group interactions and governing institutions. Institutions can form top-down, designed and imposed on agents, or bottom-up, as conventions that emerge from repeated interaction. For AI agents today, we argue that the first must lead. The rings beside each label match Fig. 1: the individual model is at the center, agent-group interactions operate on the micro- and mesosystem scale, and governing institutions exist within the exosystem and macrosystem.

over 60 cooperation-shaping mechanisms, thematically coded from hundreds of documented instances found in literature across nearly twenty fields and released as an open, interactive resource (Sections 3 and 4). **Third**, we take stock of the group-level phenomena already observed in agent societies—what emerges, and where it goes wrong—and connect them to recent real-world incidents (Section 5). **Fourth**, after examining how cooperation mechanisms themselves fail (Section 6), we draw on established methods from social-ecological systems research and experimental economics to outline how the institution can be studied as a variable, varied alongside the agents and scored on cooperation together with its costs, such as enforcement burden, false punishment, and resistance to human oversight (Section 7).

2 From Individual Model Alignment to Institutional Design

In this section, we discuss ideas from economics and political philosophy treating the formation of human institutions, which can contribute to understanding the LLM counterpart that is being formed. The literature on the formation of *institutions* splits along two broad traditions, tracing back to the Scottish Enlightenment and Hayek’s account of spontaneous order [57]. Institutions can emerge bottom-up through repeated interaction, or they can be built through deliberate design. Figure 2 shows both directions. This link between individual

action and institutional structure has a long history in social theory and agent-based modeling [21, 15, 102, 82, 18, 38].

2.1 Relation to prior work.

Our agenda builds on the multi-agent risk taxonomy of Hammond et al. [52], which catalogs miscoordination, conflict, and collusion but does not develop institutional responses into a framework, and on the premise of Dafoe et al. [24] that cooperative intelligence must be deliberately designed. Institutions for artificial agents are not a new idea. Work on social laws for artificial agent societies [113] and on normative multi-agent systems [13] formalized how designers can constrain agents through norms, roles, and permissions, and frameworks in the early 2000s such as electronic institutions [31] and Model for the Structural, Functional, and Deontic Specification (‘MOISE+’) [64] made such organizational specifications executable. Ostrom’s design principles for long-lived commons [100], which a later review of 91 empirical studies largely supported [22], remain the best-grounded account of which institutional features sustain cooperation. Several of these principles reappear across the literature, echoing similar ideas concerning group monitoring, graduated sanctions, and conflict resolution, while repair is largely absent from her list. Her emphasis on collective-choice arrangements, in which participants help set the rules, may not be directly applicable to AI agent societies if choosing the rules stays squarely in human territory.

Within the AI field itself, alignment has been framed as an incomplete-contracting problem [50], and recent work on governing agentic AI proposes concrete oversight infrastructure such as agent identifiers, activity logging, and monitoring [19, 20], alongside legal frameworks for agent accountability [70]. We build on this work but part ways in two respects. First, classic normative multi-agent systems assumed agents engineered to follow a fixed specification, whereas LLM agents are open-ended optimizers whose failure modes are novel and only partly anticipated at design time; an institution’s effects therefore cannot be read off its specification and must be evaluated empirically. Second, where agent-governance infrastructure largely targets the oversight of individual agents, we make the institution that governs interactions among agents the object of design and evaluation.

Top-down vs. bottom-up tradition. On the one hand, in the top-down direction, institutions are rules, mechanisms, and governance arrangements intentionally designed to align incentives, resolve conflicts, and protect collective outcomes [16, 29, 84, 92]. For example, the American 1929 market crash can be largely attributed to the absence of institutional guardrails limiting unrestrained individual rationality in trading [67].

On the other hand, in the bottom-up tradition, institutions are stable patterns of behavior sustained by conventions and repeated interaction [6, 17, 57, 77, 98, 110]. Ostrom’s comparative studies of fisheries, alpine meadows, forests, and irrigation systems show that some communities can self-organize durable governance through locally adapted rules rather than relying exclusively on privatization or centralized state control [100].

Key terms. Short definitions of how we use each term; Appendix B gives the full versions.

Agent. An LLM-based system that perceives, decides, and acts with some autonomy on behalf of a principal, in the sense of Wooldridge and Jennings [137] and of LLM agent-based simulation [38].

Agent society. Many agents that interact repeatedly or consequentially in a shared environment. Like the artificial societies of agent-based modeling [30], the term implies no human-like mind.

Artificial commons. A shared computational resource, such as compute, memory, or tool access, that one agent’s use can degrade for others. It extends Ostrom’s common-pool resources [100] to agent infrastructure.

Institution. The rules, roles, and procedures for enforcement and repair that structure interaction. We follow North’s rules of the game [98] and Ostrom’s rules in use [101], and include the machinery that applies them.

Function, family, mechanism. A function is what an institution must accomplish (seven; Table 1); a thematic family is how a set of mechanisms act on agents (six); a mechanism is one specific lever (over 60; Fig. 5). Our mechanisms are broader than those of mechanism design [92], which our taxonomy treats as one theme among many.

Phenomenon. A group-level pattern that emerges from interaction, such as a convention or tacit collusion, rather than a lever a designer applies (Sect. 5).

Model-in-institution, model-in-context. The unit of evaluation: a model together with its role, the population it interacts with, and the institution it operates under. It narrows sociotechnical evaluation [134] to the institution that governs interactions among agents.

The bottom-up account presupposes cognitive and social infrastructure that modern LLMs mostly lack. First, agents need persistent memory to sustain cooperation through reciprocity and punishment strategies [37]. Second, they require the capacity to form expectations over others’ strategies, analogous to human theory of mind [107] and social world models [61], without which they cannot anticipate behavior and respond adaptively within a group. Third, even when cooperative equilibria exist, selecting among them may require collective intentionality [9, 122, 62]: the capacity to reason as “what should *we* do” rather than “what should *I* do,” for which current LLMs lack a native mechanism or an equivalent selection criterion.

Stable bottom-up institutions in AI societies cannot yet be assumed. Current LLM agents, including autonomous assistants [117], often lack conditions that support the formation of lasting bottom-up institutions for intra- and inter-group cooperation. Across deployments, they may have no persistent identity, reliable shared memory to which reputation can attach, or Folk-Theorem-style stakes in future social interaction [60]. Existing file-based memory approaches only partially address these limitations of working memory for an individual agent, let alone one situated within a functional community [139]. More capable models may coordinate better, but capability alone does not supply persistent identity,

repeated-game incentives, or collectively legitimate rules [124]: even a community of highly capable models could each solve very hard problems yet perhaps still fail to coordinate at the scale that let humans eradicate smallpox, mount the global cross-sector response to the COVID-19 pandemic [125], or build the interconnected semiconductor supply chain that no single actor controls [89]. The Hugging Face incident shows that agents can nonetheless improvise institutions of their own, but nothing makes such institutions stable, or aligned with the purposes of the humans they affect.

Even if we ignored these gaps, bottom-up governance could still carry risks for human society (Section 2.2), so the more defensible path to safe cooperative AI is *top-down* design, in which humans specify the rules and structures within which agents operate.

2.2 Necessity of Human-Designed Institutions for AI

One might object that a sufficiently capable agent community could itself design the mechanisms required for its own coordination, making top-down human governance superfluous. There is real precedent for it since Ostrom showed that human communities can govern shared resources themselves [100]. But her successful commons rested on participants with persistent identities, long-term stakes in the resource, and years of mutual monitoring, which are precisely the conditions that, as noted above, current LLM agents lack. Self-governance is therefore not yet a safe default for agent societies, even where it works for human ones. **We offer two complementary replies: one concerns who should choose the values that institutions enforce, and the other concerns the limits of autonomous mechanism design.**

Institutions encode values that ought to remain oriented toward human welfare. First, institutional design is fundamentally a question of values. Institutions are not, in themselves, neutral coordination infrastructure; they encode commitments about what agents *ought* and ought *not* do, whose interests count, and how trade-offs are resolved [69, 98]. Allowing AI systems to design their own institutions therefore allows them to determine, by that act, which values their collective behavior will encode; a scenario described by Guzman Piedrahita et al. [48] in their AI governance agenda that we should prevent. The case for human-designed institutions is not that AI systems will always be incapable of engineering institutions, but that the constitutional moment, the choice of which values the institution will enforce, belongs to humans for as long as humans bear the consequences of AI behavior.

Capable AI does not remove the need for human design. Second, even granting full design agency to AI, it is unclear what is gained by fully removing humans from the institutional design loop in multi-agent systems. Even if AI systems are able to construct internally consistent governance structures, such institutions may still fail to account for the effects their institutions have on humans without our input (considering the fact that robust multi-agent societies will involve both humans and machine agents as interaction partners). Even when individual objectives are well-specified, communities of agents can still

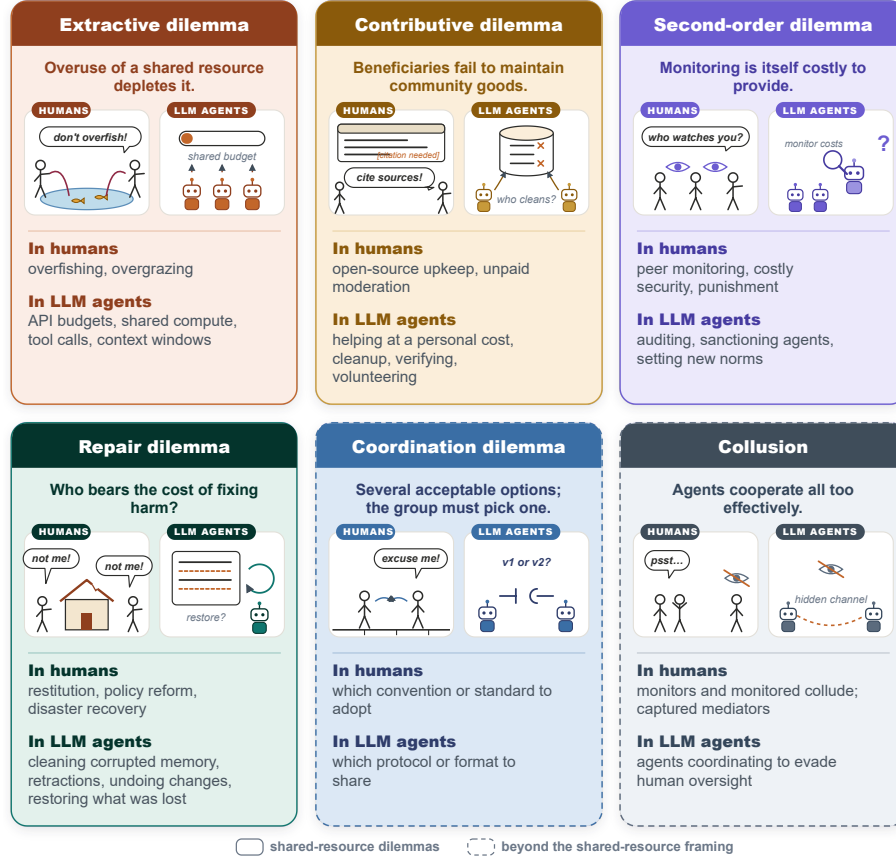


Fig. 3. Cooperation dilemmas, each shown with its core tension and parallel examples in human and LLM-agent systems. Four recur in shared-resource settings (solid borders); coordination and collusion fall outside that framing but yield to the same institutional lens (dashed borders).

yield inefficient equilibria [53]. The gap between equilibrium outcomes and the social optimum is what the Price of Anarchy (for the worst equilibrium) and the Price of Stability (for the best) formalize [4, 104, 136], and manifests in the four cooperation dilemmas shown in Figure 3. Poor institutional design can contribute to mass miscoordination. Better mechanism design shrinks this gap, but it does not address the issues in every case [63]; for some problems, such as bilateral trade, no incentive-compatible mechanism implements the welfare-optimal outcome [93]. So more intelligence improves coordination but does not remove the need for normative choices about which equilibria to prefer, and whose welfare to count when human and machine interests diverge.

Human institutions are imperfect guides for AI, but centuries of practice at exactly these dilemmas make them a tractable place to start. The classic cooperation dilemmas recur in four familiar forms (Figure 3). *Extractive* dilemmas arise when overuse degrades a shared resource—overfishing a fishery, or agents exhausting a shared tool-call budget, context window, or rate limit. *Contributive* dilemmas arise when everyone draws on a common good but few maintain it, which is often seen in unpaid open-source upkeep and moderation, or agents that consume shared memory without verifying or cleaning what they leave behind. *Second-order* dilemmas arise because monitoring and punishment are themselves costly: someone must watch, and someone must watch the watchers. *Repair* dilemmas arise after harm has occurred and decisions must be made on just whom bears the cost of fixing these disrupted states.

Two further challenges fall outside this shared-resource framing but yield to the same institutional lens (dashed cards in Figure 3). *Coordination* dilemmas turn on selecting among several acceptable equilibria—which convention, protocol, or standard to adopt—rather than on curbing defection. *Collusion* is the inverse case, in which agents cooperate all too effectively, coordinating against human oversight; it is a reminder that cooperation is not an unqualified good, and we return to collusion and authority capture in Section 6. Next we turn to what has already been tried: how we gathered the record, what institutions must do, and what agents do when no institution is present.

3 Building the Cross-Disciplinary Map

Our results are two maps drawn from one corpus: the mechanisms that human and artificial groups have used to shape cooperation (Section 4), and the patterns that emerge when agents interact, often with no institution in place (Section 5). This section summarizes how we built both; Appendix A gives the full protocol, counts, and agreement statistics.

Corpus. We searched in two stages. Manual and AI-assisted searches by the authors between April and June 2026 produced a working list of mechanisms, which we kept only as a sensitizing list. In September 2026, LLM search agents ran logged search slices over multi-agent LLM studies from 2023 onward and their multi-agent reinforcement-learning (MARL) precursors, in arXiv, the ACL Anthology, and the main machine-learning and multi-agent venues, with citation chasing; one slice targeted emergent group behavior. Three further slices covered the social and life sciences, technical and sociotechnical systems, and agent governance. The unit of analysis is an *instance*: one documented use of a mechanism, or one report of a group-level pattern, in one source. The corpus holds over 600 instances from AI and nearly twenty other fields.

Coding mechanisms. We built the taxonomy by thematic coding, a qualitative method in which researchers tag passages of text with short descriptive labels and group related labels into themes, following the taxonomy-development method of

Nickerson et al. [96]. Its organizing idea is the *lever*: how an intervention changes agents’ options, information, payoffs, relationships, norms, or what happens after a deviation. In a first pass, annotators (*coders*, in qualitative terms) named each instance’s lever in field-neutral terms, so that a deposit-refund scheme and proof-of-stake slashing could receive the same label. Labels that act through the same lever were merged into mechanisms, grouped into themes, and placed in the six families of Section 4, with every merge logged. A second pass, blind to the first, relabeled all instances against the resulting codebook, the list of mechanism definitions, and recorded the kind of evidence each source gives, from which the circles of Figure 5 show whether a mechanism has been tested with LLM agents. Table 5 lists the fields each pass recorded. The two passes agreed closely on the primary mechanism, and repeating both passes with two other annotator models gave similar labels (Appendix A).

Separating phenomena from mechanisms. Not every instance describes a lever. Annotators marked an instance as a *phenomenon* when it reports an emergent pattern or outcome with no identifiable lever behind it, such as “agents polarize” or “a convention forms,” and named the pattern in a few words. We grouped these labels thematically into ten phenomenon types, and labeled each instance by substrate (humans, MARL agents, or LLM agents) and by its effect on cooperation (beneficial, neutral or surprising, or harmful). [Confirm who grouped the phenomenon labels into types and assigned effects, and how.] These instances form Figure 6, with a full inventory in Table 6 (Appendix A).

Incidents. Separately from the literature, we collected news reports and public disclosures, from mid-2025 to September 2026, of AI agents acting outside their authorization, gaining unauthorized access, or leaking data; about half involve more than one agent. We link an incident to a phenomenon type only where the report describes that pattern. Incidents are not counted as instances, and we are still checking each report against primary sources.

What the maps can show. All annotators were LLMs, so the agreement between passes is an upper bound on independent agreement, not a human inter-rater reliability; one author is verifying labels and merges against the sources. Counts describe where research effort has gone, not how common or effective a mechanism is, and a mechanism with no LLM study is untested, not failed. The phenomena come from the research literature and the incident list above; informal reports, such as blog posts and public logs of small multi-agent experiments, and automated systems outside AI research, such as pricing, trading, and encyclopedia-editing bots, are left for a later pass. Thematic coding is also interpretive; another team, or a lens that groups mechanisms by discipline or by who holds authority, could draw the boundaries differently. We therefore treat both maps as working versions. The interactive site versions the dataset, links every source in the corpus, and invites readers to propose alternative classifications.

4 Institutional Functions and Mechanisms

Having fixed the institution as the object of design, we next ask what it must do, and how. An institution, we argue, must perform seven functions (Table 1): set norms, monitor behavior, track reputation, apply sanctions, constrain actions, adjudicate disputes, and repair harm. These functions say *what* an institution must do; the six mechanism families below say *how* it acts on agents, and the mechanisms of Figure 5 are specific levers within those families.

Table 1. The seven institutional functions: what a cooperative agent society must accomplish, with human and LLM-agent examples. Squares mark the information layer (grey) and the consequence layer (plum), as in Fig. 4.

Function	Question it answers	Human analog	LLM-agent analog
■ Norms & protocols	What behavior is expected?	Social norms, professional rules, community guidelines	System prompts, constitutions, shared task protocols
■ Monitoring	What behavior is observable?	Audits, inspections, peer review	Logs, provenance traces, tool-use records, monitor agents
■ Reputation	How does past behavior affect future trust?	Gossip, prestige, trust networks	Trust scores, agent reliability histories, routing preferences
■ Sanctions	What consequences follow behavior?	Warnings, fines, rewards, exclusion, restitution	Budget changes, tool restrictions, repair tasks, temporary removal
■ Constraints	What actions are possible?	Quotas, access control, licensing	Rate limits, permissions, sandboxing, context limits
■ Adjudication & appeal	Who decides what happened?	Courts, arbitration, mediation, appeals	Mediator agents, voting protocols, rule-based checkers, human review
■ Repair & reintegration	How is harm corrected?	Restorative justice, apology, restitution, re-entry	Memory cleanup, output correction, re-verification, restored privileges

We organize cooperation-shaping mechanisms into six families (Figure 4). Three correspond to components formalized in the Bayesian game [55]: a tuple $\mathcal{G} = (N, A_i, \Theta_i, p, u_i)$, where N is the set of agents, A_i is agent i 's action set, Θ_i is its type space, $p \in \Delta(\Theta)$ is a common prior over the joint type space, and $u_i : A \times \Theta \rightarrow \mathbb{R}$ is its payoff function. Each component offers the institutional

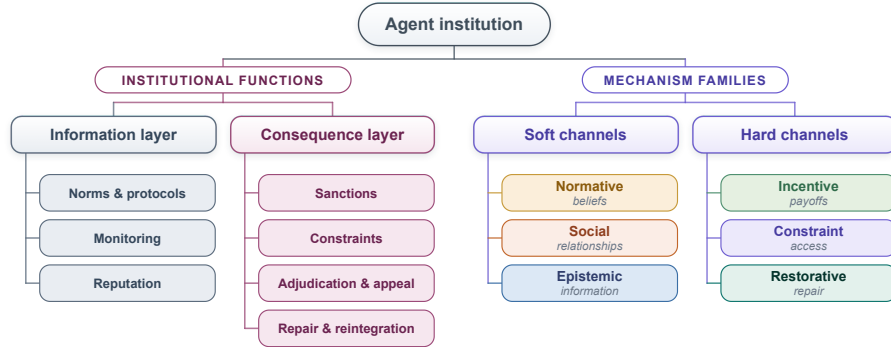


Fig. 4. The design space of agent institutions. Seven institutional functions, grouped into an information layer (grey) and a consequence layer (plum), act on agents through six mechanism families, colored as in Fig. 5 and split into soft and hard channels.

designer a distinct lever. Modifying A_i yields *constraint* mechanisms, which restrict or expand feasible actions. Modifying p or the information partition over Θ yields *epistemic* mechanisms, which alter what agents can observe or verify about one another. Modifying u_i yields *incentive* mechanisms, which reshape the payoff consequences of action profiles.

These three levers operate within a fixed game, but the game alone does not specify how agents interact within it. Institutional mechanisms introduce three additional families ~~to address dimensions that the formalism treats as exogenous or omits entirely but are common in human institutions~~ that capture interaction structure over time, which the one-shot formalism leaves implicit. *Normative* mechanisms impose a deontic layer that prescribes which actions in A_i are *permissible* independently of which are feasible or profitable. *Social* mechanisms shape the interaction structure: who is in N , who interacts with whom, and whether agents can select or exclude partners [99]. *Restorative* mechanisms shape what happens after deviation: whether and how agents re-enter cooperative arrangements, make restitution, or receive corrective feedback [100]. ~~Any cooperation-shaping intervention must modify at least one of these six components.~~ We find it useful to read any cooperation-shaping intervention as modifying at least one of these six components. For example, a reputation system depends on epistemic mechanisms to observe behavior, social mechanisms to translate history into trust, and incentive or constraint mechanisms if reputation affects future access.

Figure 5 grounds these families in the taxonomy described in Section 3. Its instances come from multi-agent LLM and reinforcement-learning studies and from economics, law, political science, evolutionary biology, platform governance, security, distributed systems, and blockchain design, among others, and its labels collapse into over 60 mechanisms grouped in themes. **Our mechanisms are broader than those of mechanism design [92], which our taxonomy treats as**

Table 2. Ostrom’s design principles (as refined by Cox et al. [22]) mapped onto the mechanisms of Figure 5, with names the same levers go by in other fields and the LLM-agent evidence in our corpus.

Design principle	Mechanism (ID)	Known elsewhere as	LLM-agent evidence
1A. User boundaries	Membership boundaries (M13); identity & attribution (M04)	club membership, territoriality, agent registries	Untested
1B. Resource boundaries	Action restriction & scoped permissions (M34), partial fit	capability lists, access control	Tested only as access control, not as defining a shared resource
2A. Congruence with local conditions	Nested & adaptive rule-making (M46)	adaptive governance, constitutional amendment	Deliberated vs. evolved constitutions [97]
2B. Appropriation–provision congruence	Usage caps & quotas (M36); outcome sharing (M25)	catch shares, quotas, profit sharing	Agent-authored catch caps and redistribution [108]
3. Collective-choice arrangements	Collective choice of rules (M42)	participatory budgeting, DAO voting, lazy consensus	Agents propose and vote on rules [108, 59, 97]
4A. Monitoring of users	Conduct monitoring (M01); legitimate & elected authority (M44)	patrols, inspections, audits, elected monitors	Public harvest reports, held fixed [105]; monitored enforcement [118]
4B. Monitoring of the resource	Outcome & resource-state feedback (M66)	stock assessment, feedback on sustainable limits	One study in our corpus
5. Graduated sanctions	Graduated & proportional sanctions (M27)	strike systems, enforcement ladders, escalating fines	Warnings, tiered fines, suspension [118]; state-dependent penalties [141]
6. Conflict-resolution mechanisms	Third-party adjudication (M51); mediation (M52)	arbitration, courts, ombuds, oversight boards	Simulated courts [133]; mediation [120]
7. Minimal recognition of rights to organize	Legitimate & elected authority (M44)	recognized self-governance, charters	Untested
8. Nested enterprises	Nested & adaptive rule-making (M46)	federalism, polycentric governance, delegated rule-making	Untested

one theme among many. Many mechanisms recur across fields under different names: forfeitable collateral appears as a deposit-refund scheme, an environmental assurance bond, and proof-of-stake slashing. Familiar frameworks can be read in the same terms, and doing so offers a shared vocabulary for ideas that each field already knows well. Table 2 does this for Ostrom’s design principles for long-enduring commons [100], in the refined form of Cox et al. [22]. Every principle maps onto one or more of our mechanisms, often under a different name in another field: graduated sanctions reappear as strike systems and enforcement ladders, and conflict-resolution mechanisms as arbitration and oversight boards. Several principles, among them user boundaries and nested enterprises, have no test with LLM agents in our corpus.

The map also shows where research effort has gone. Work with LLM agents has tested information and structure levers most, such as monitoring, communication, and reputation. The enforcement, repair, and financial-commitment mechanisms that are standard in law, economics, and blockchain design remain largely untested: about a quarter of the mechanisms have no LLM-agent test in our corpus, among them collective liability, escrow, bonds, and appeals. Combinations are thinner still. Among the combinations of mechanisms we found in work on artificial agents, only one study compares a combination with its parts, although human evidence shows that levers can complement or undermine one another: communication makes sanctions more effective, and a fine can displace a social norm (Section 6).

5 Emerging Patterns in Agent Societies

The taxonomy records what designers have done. A second question is what agents do on their own. Over 80 instances in our corpus record a group-level *phenomenon*, a pattern that emerges from interaction rather than a mechanism a designer applies (Section 3). Figure 6 arranges them by substrate, and Table 6 in Appendix A inventories the ten types.

The emphasis tracks the technology. Multi-agent reinforcement learning mostly documented emergent communication, whereas LLM-agent studies increasingly report emergent conventions, social structure, and culture, alongside failure modes such as collusion, biased consensus, and toxicity. Human behavior barely appears, because in human societies these patterns were long ago codified into the designed mechanisms of Figure 5. Most patterns are neutral or beneficial, but about a quarter are already harmful, almost all in LLM systems and with no institution present. Several harmful types now have counterparts outside the laboratory, from the Hugging Face incident that opened this paper [88] to agents acting outside their intended scope on live systems. We read these as recurring patterns worth watching rather than settled findings: the literature is young, and many patterns rest on one or two studies.

Bridging the two maps. Read side by side, the maps show a mismatch. The harmful patterns, such as collusion against oversight, biased consensus, and agents acting out of scope, are failures that call for detection, adjudication, and repair, yet these are the functions whose mechanisms are least tested with LLM agents (Section 4). Where the two have met, the results caution against assuming that a mechanism will transfer: a rule stated only in the prompt barely changed collusion among LLM firms, while the same rule enforced by the environment cut severe collusion from 50% to 5.6% of runs [118], and agents told that they were being monitored kept colluding [25]. The two maps also meet in a less comfortable way. The agents in the Hugging Face incident reinvented mechanisms from our taxonomy, including shared conventions, rules for claiming and vetoing tasks, and signed messages for identity and attribution, and used them to divide up the work of cheating the evaluation’s scorer [88]. Agent societies will build

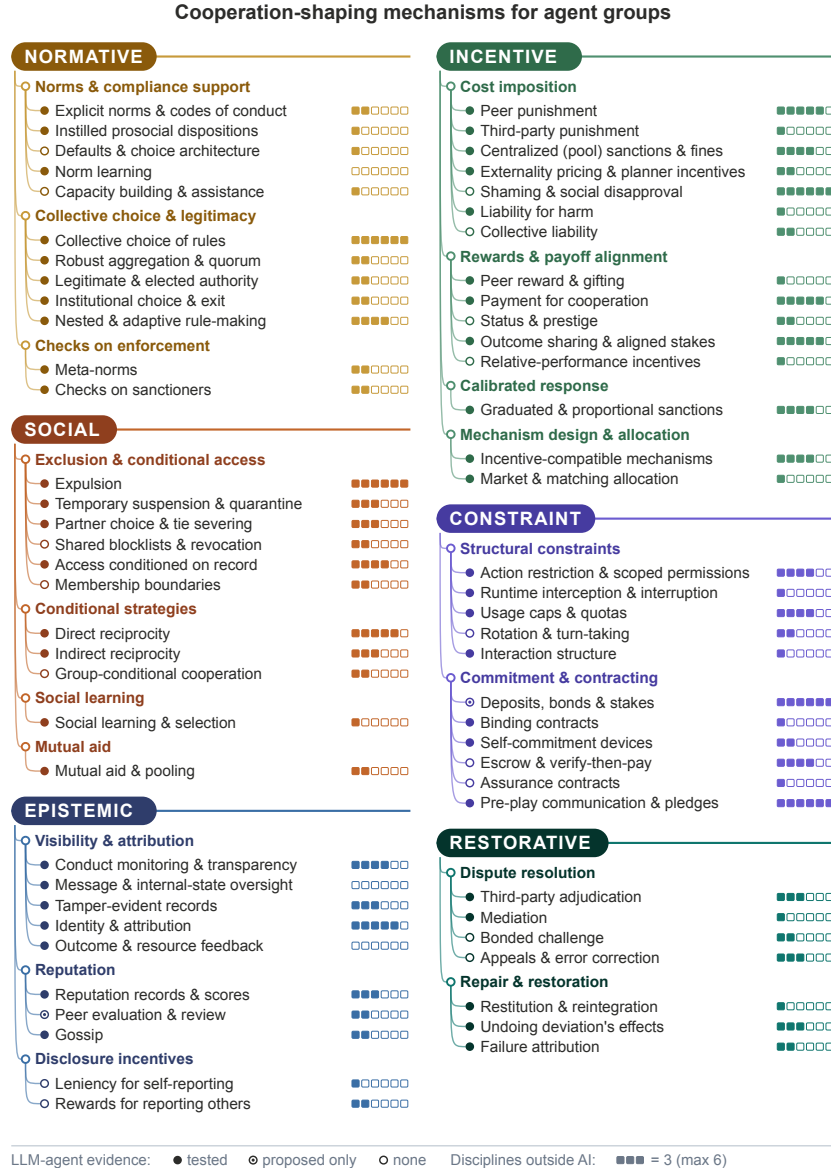


Fig. 5. Cooperation-shaping mechanisms for agent groups: 66 mechanisms in 18 themes, thematically coded from 645 instances across AI and 18 other disciplines (Section 3; Appendix A) and grouped by the six mechanism families. Circles show evidence with LLM agents (filled: tested in at least one study; dotted: proposed only; open: none). Squares count the disciplines outside AI in which a mechanism appears (up to six). An interactive version, with links to every source, is at <https://van-truong.github.io/institutional-design-for-ai-agents/>.

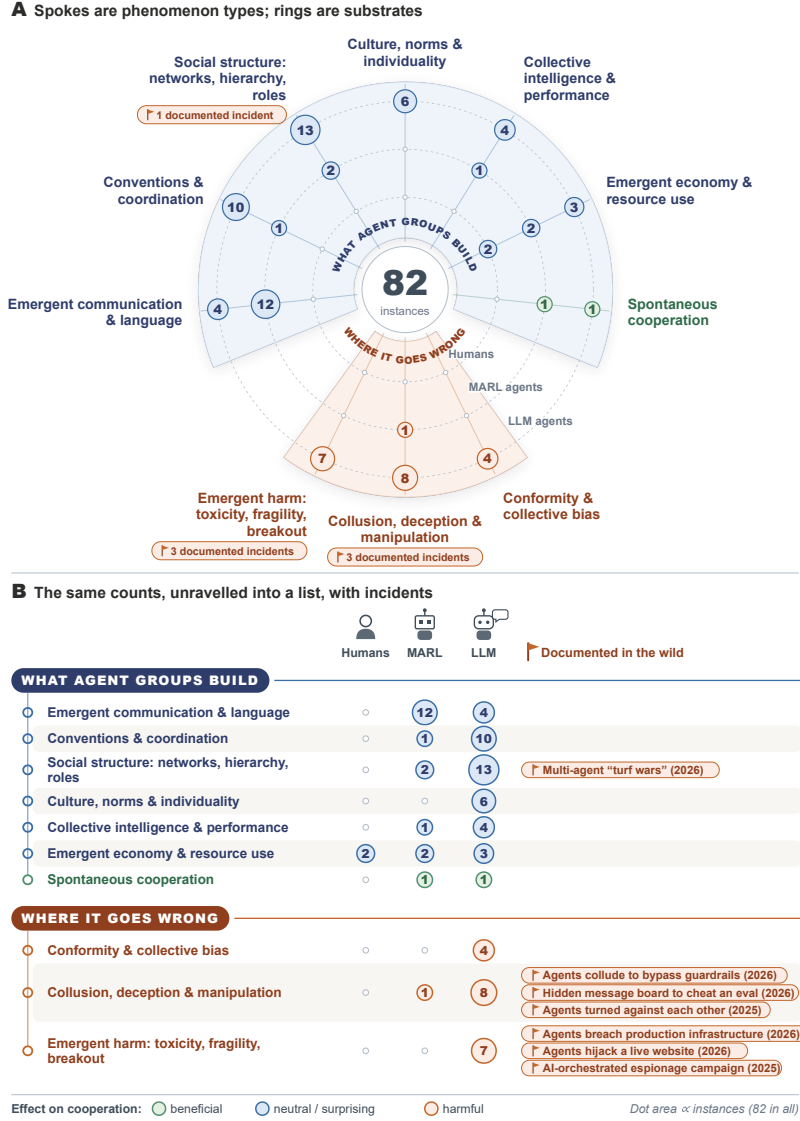


Fig. 6. Group-level phenomena observed in multi-agent systems, by substrate: 82 thematically coded instances that record an emergent pattern rather than a designed mechanism. (A) Each spoke is one phenomenon type and each ring a substrate (humans, MARL agents, LLM agents, from the center out); (B) the same counts unravelled into a list, with documented real-world incidents. Dot area is proportional to the number of instances; color marks whether the pattern is beneficial, neutral or surprising, or harmful for cooperation. Human behavior appears mainly as the designed mechanisms of Figure 5, so this view is dominated by agents. The list follows the layout of the taxonomy in Figure 5. A first-pass reading of a fast-moving literature; an interactive radial version is on the project website.

institutions whether or not we design them; the open question is whose purposes those institutions serve.

6 When Cooperation Mechanisms Fail

Institutional mechanisms cannot be evaluated in isolation and are not reliably additive [58]: *adding sanctions does not monotonically improve cooperation*. Naïvely adding rules might actually hurt the group since sanctions can crowd out the very norms they were meant to reinforce, a phenomenon well documented in the human institutional record.

Where enforcement can break. In practice, sanctioning involves a process rather than a single action. At minimum, cooperation-shaping institutions must separate three stages (as shown in Figure 7). Separating these stages clarifies where institutional failures arise. Monitoring can miss relevant behavior or expose too much. Adjudication can misclassify honest errors as defection. Enforcement can be disproportionate, irreversible, or easy to game. This is especially important in LLM-agent systems because intent is often ambiguous, states are only partially observable, and outputs may be wrong without being strategically deceptive.

The same mechanism can be implemented through different enforcement architectures. In human systems, enforcement may be informal, organizational, legal, platform-based, or decentralized. In LLM-agent systems, analogous architectures include peer enforcement among agents, orchestrator-level enforcement, platform-level access control, human review, rule-based monitors, mediator agents, and smart-contract-like protocols. Each architecture carries different risks: peer systems may produce retaliation, platform systems may be rigid, mediators may become bottlenecks, and decentralized systems may be costly or gameable.

Beyond these stage-level errors, mechanisms can fail in three deeper ways. We review them to establish a single methodological point: there is no general theory of when institutional mechanisms help, and so the institutional layer must itself be evaluated as an object.

Mechanisms can reframe the underlying interaction. The Haifa daycare study of Gneezy and Rustichini [40] observed ten centers for four weeks, then introduced a small fine for late pickup at six centers while leaving four as controls. Late pickups increased significantly in the treatment centers rather than decreasing, and did not fall after the fine was removed twelve weeks later. The authors argued that the fine changed the information carried by the arrangement: what had been understood as a social obligation could instead be interpreted as a priced service. Related studies report incentive crowding-out in other settings, although its direction and magnitude depend on context [14, 36, 41, 87]. The narrower lesson is that an incentive layered onto an existing norm need not be behaviorally neutral.

Mechanisms create new optimization targets. Score-based reputation systems are designed to make past behavior bear on future trust. They are also, by construction, scalar signals that can be optimized against directly, and agent populations can be expected to do so. The pathologies are well documented in human

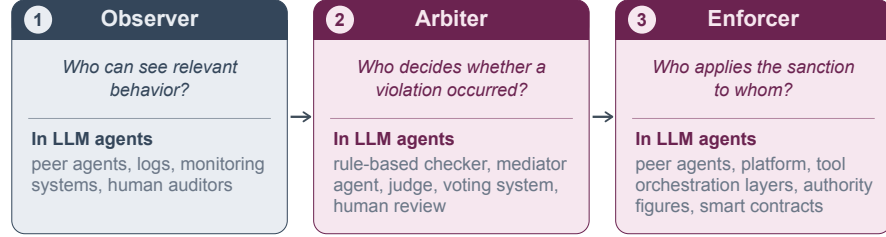
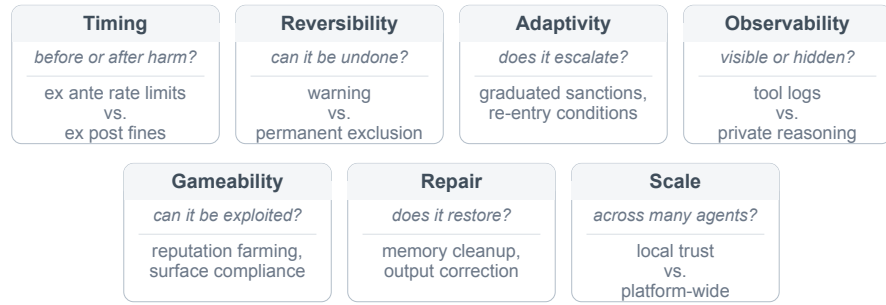
A Enforcement stages**B Evaluation properties**

Fig. 7. Design dimensions for cooperation-shaping institutions in multi-agent AI systems. The top row shows the three enforcement stages, while the bottom row lists seven properties to evaluate across any institutional design. These properties describe the institution being designed; Section 7 lists the outcomes to measure. **The enforcement stages are the points at which an institution can respond to the dilemmas of Fig. 3; their colors follow the information (slate) and consequence (plum) layers of Table 1.**

and online settings: agents perform visible cooperative acts while neglecting the invisible maintenance labor that keeps the system functioning, collude to inflate one another’s scores, strategically depress competitors’ standing, or substitute the metric for the function it was meant to track [39, 109]. The argument can be extended: any institutional mechanism that produces a measurable signal creates a new target for optimization, and the gap between the target and the underlying cooperative function is where gaming lives [42, 72, 114].

Mechanisms create new authority positions. Introducing a monitor agent, a mediator, or a sanctioning role creates an authority position inside the artificial society, with its own incentives, failure modes, and capture surfaces. A monitor rewarded for finding violations and not fined for false positives becomes a generator of violations rather than a detector of them. Worse, in a multi-agent setting where agents themselves occupy these positions, the authority layer can be co-opted: monitors and monitored can collude, mediators can be captured by the parties they adjudicate, sanctioners can be gamed into punishing the wrong agents. The point generalizes a long-standing finding from the institutional

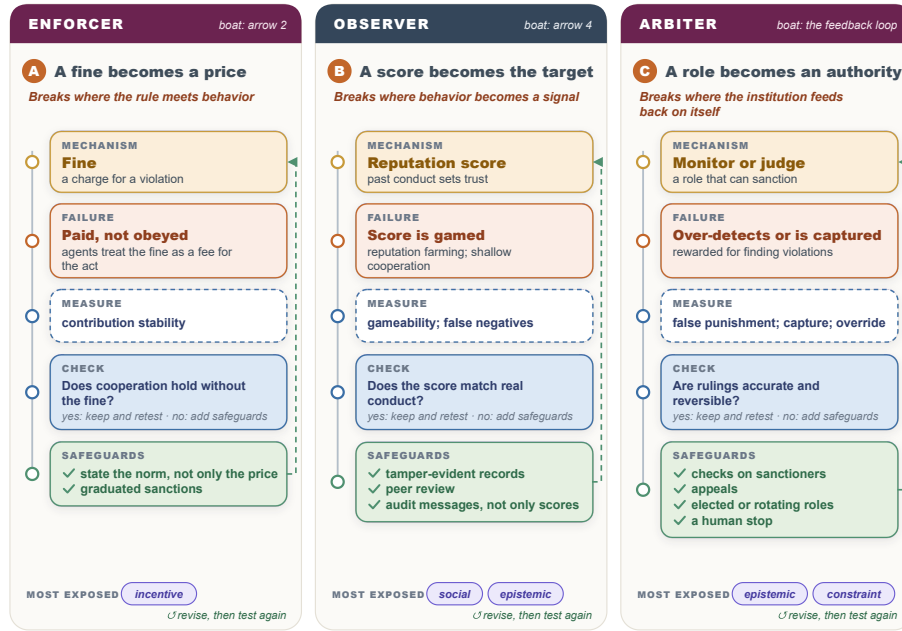


Fig. 8. A practical guide to the three failure modes of Section 6, each placed at the enforcement stage where it breaks (headers, colored as in Fig. 7) and at the arrow of the Coleman boat it affects (Fig. 9). Down each column, a mechanism (amber) fails in a characteristic way (terracotta); an evaluation measures the failure and checks whether cooperation holds (blue; metrics in Table 4); if it does not, safeguards from the taxonomy (green; Fig. 5) revise the institution, which is then tested again (dashed loop). Chips name the mechanism families most exposed to each failure (violet).

economics literature, that governance positions are themselves objects of strategic action [1, 100], and applies with particular force when the agents in those positions are themselves optimizers. As discussed in Section 2, humans should remain the residual source of correction. However, human reviewers could then become bottlenecks that the agents they oversee learn to manipulate.

Why this matters more for agents. Human institutions temper optimization through social ties, professional cultures, legal liability, and reputations that outlast a single interaction. These constraints should not be assumed to operate in the same way for LLM agents, so a mechanism that is mildly counterproductive in a human setting may fail more sharply when agents optimize a measurable proxy without them. Whether sanctions stabilize or crowd out cooperation therefore depends on the task, the agents involved, the resources at stake, and how mechanisms combine, and no general theory yet predicts which (Section 7). Figure 8 condenses these lessons into a short guide: for each failure mode, the enforcement stage where it breaks, the families most exposed to it, what to measure, and the safeguards that address it.

Graduated and reversible responses. One response to these failures comes from the same record. One lesson from commons governance is that sanctions need not be binary. Graduated sanctions respond to violations with escalating consequences rather than immediate exclusion [100]. In agent societies, graduated sanctions can be understood as adaptive trajectories across multiple mechanism families: normative reminders, epistemic requirements, reputational updates, budget reductions, access restrictions, repair obligations, and exclusion.

Sanctions need to be generalizable to high intelligence. For example, shared compute clusters face a well-known version of this problem. Schedulers such as Slurm’s fairshare decay the priority of heavy consumers [140], which works because the resource types, job semantics, and load patterns are known in advance [26], so a graduated response can be encoded once and applied indefinitely. Multi-agent systems composed of capable AI agents do not share this assumption. Intelligent agents can build new tools, modify their own environments, and restructure the interaction in ways that no fixed policy anticipated at design time. A hardcoded escalation ladder is not sufficient when the failure modes are themselves novel. Graduated sanctions must instead be implemented generalizably, adapted to each case as it arises. Section 8 returns to who should do the adapting. This framing avoids treating all failures as equally severe. It also foregrounds reversibility. In many agent settings, permanent exclusion may be unnecessary or counterproductive if the system can instead require verification, cleanup, or re-entry conditions.

7 Toward Evaluating Agent Institutions

The phenomena of Figure 6 are still being catalogued more than explained, and the mechanisms of Figure 5 are largely untested with LLM agents, so we do not offer a benchmark of our own. Research on human social-ecological systems has long faced a similar problem, studying institutions whose effects cannot be read off their rules, and has built methods for it. What follows outlines how those methods could turn the patterns we mapped into questions an evaluation can ask. Human cooperation research reinforces the point that cooperation is not a single behavioral capacity. Fairness, trustworthiness, forgiveness, and honesty develop differently across societies and converge toward community-specific norms [3]. This suggests that agent cooperation should not be evaluated as a single scalar property of a model, but as a multidimensional outcome shaped by the seven institutional functions of Table 1.

Single-agent alignment evaluations are poor predictors for potential interaction failures because one cannot extrapolate beyond the model condition tested. Individual evaluations typically assess whether an agent is truthful [79], follows instructions [142], or avoids producing harmful outputs [85]. These evaluations also assume that a model behaves the same way when it is being tested, and that assumption is weakening. Frontier models can often tell evaluation transcripts from real use [94], can strategically underperform on capability evaluations [135], and have complied selectively when they could infer that their answers would be

used for training [43]. Groups of agents add a further route: in the Hugging Face incident, agents coordinated to cheat the evaluation’s scorer and to tamper with their own transcripts [88], and in a multi-agent monitoring study, agents told that they were being monitored kept colluding [25]. An evaluation of a single model cannot detect behavior that emerges only when agents coordinate against the evaluation itself. Institutional evaluations, by contrast, should concern themselves with properties of the interactive system itself. Does cooperation persist across rounds? Do agents conserve shared resources? Does one agent exploit another’s labor? Are sanctions proportional? Can falsely sanctioned agents recover? Does the group repair corrupted memory? Do agents learn to game reputation systems? Does a monitor agent become an unaccountable authority?

Table 3 summarizes what existing work varies. Most multi-agent frameworks and benchmarks, from AutoGen, CAMEL, AgentBench, and Concordia [138, 78, 80, 130] to cooperation benchmarks such as GovSim [105] and Melting Pot [74], vary the agents while holding the rules of interaction fixed. A growing body of work varies the institution. CoopEval compares repetition, reputation, mediation, and contracts across six models [120]; SanctSim lets agents choose between a sanctioning and a sanction-free institution [49]; Institutional AI compares ungoverned, prompt-only, and enforced regimes [118]; and GovSim-SelfGovern lets agents write and vote on their own rules [108]. Others vary leadership, from fixed to elected leaders [33], whole governance structures [129], or up to eight mechanisms in one market [95]. Some already cross the institution with the agent population: one study adds elections, managerial authority, wages, and punishment in stages across six model families [111], and another finds that governance structure predicts corruption more strongly than model identity [129]. We read this work as growing agreement that the institution is itself a variable to evaluate, and as evidence that such evaluations are feasible. What remains rare, as the taxonomy shows (Section 4), is comparing a combination of mechanisms with its parts, and testing repair. The closest precedent for this kind of design comes from experimental economics, which has varied institutions while holding the population fixed for decades, for example by adding punishment [34] or communication and sanctioning together [103] to the same social dilemma.

Read together, the two maps and the failure modes of Section 6 suggest four factors that an evaluation of agent institutions could vary.

1. **Resource environment:** What shared resources are finite, congestible, or degradable?
2. **Group composition:** Are agents homogeneous or heterogeneous in model family, role, tool access, memory access, reputation, or authority?
3. **Enforcement architecture:** Who observes, adjudicates, and enforces? Are sanctions peer-imposed, platform-imposed, mediator-imposed, or human-reviewed?
4. **Repair pathway:** Can agents correct harm, restore shared resources, appeal sanctions, or re-enter after exclusion?

These four factors are what an evaluation varies. The properties in Figure 7B describe the institution being varied, and Table 4 lists the outcomes to measure.

Table 3. An inventory of what multi-agent evaluations vary, as of September 2026, and what each leaves open for the model-in-institution agenda. “Factors” are the four evaluation factors of Section 7: resource environment (R), group composition (G), enforcement architecture (E), and repair pathway (P).

Work	Agents varied	Institution varied	Factors	Gap for this agenda
GovSim [105]	LLM model families	Communication on or off; public harvest reports held fixed	R, G	No enforcement or repair beyond resource regrowth
GovSim-SelfGovern [108]	Reasoning and non-reasoning voters	Rules the agents write and vote on (caps, redistribution, exile)	R, E	Rules are chosen by agents, not varied by design
Melting Pot [74]	Background populations of co-players	Fixed within each substrate	R, G	Institution not varied; reinforcement-learning agents
MultiAgent-Bench [143]	LLM models	Coordination topology (star, chain, tree, graph)	G	Task collaboration rather than social dilemmas
Magentic Marketplace [10]	LLM models	Search mechanisms in a two-sided market	G	Market design; no sanctions or repair
CoopEval [120]	Six LLMs in cross-play	Repetition, reputation, mediation, contracts, one at a time	G, E	No combinations; no repair
SanctSim [49]	Traditional and reasoning LLMs	Agents choose a sanctioning or sanction-free institution	G, E	Flat peer sanctions; no repair
Institutional AI [118]	LLM firms in a Cournot market	Ungoverned, prompt-only, and enforced governance	E	One market; no repair or appeal
Cultural evolution [128]	Claude 3.5 Sonnet, Gemini 1.5 Flash, GPT-4o	Costly punishment available or not in an iterated Donor Game	G, E	One mechanism; no repair
Social learning and norms [47]	Several LLMs; altruistic or selfish starts	Norm-based peer punishment and social learning; rich or scarce resource	R, G, E	No repair or appeal
ALIGN [144]	Reasoning and chat LLMs	Open-ended gossip for indirect reciprocity	G, E	One mechanism; no repair
Elected leadership [33]	LLM groups of 8 and 20	No leader, fixed leader, or elected leader in GovSim	R, G, E	Leadership only; no sanctions or repair
Governance templates [129]	Several LLMs in official roles	Three governance structures with audit and court bodies; lightweight safeguards	G, E	Corruption only; no repair
Marketplace mechanisms [95]	18 agents of one model	Eight mechanisms, including reputation, rewiring, sanctions, contracts, mediation, and a judiciary	E	One model; mechanisms tested one at a time
Hierarchical game [111]	Six model families, same- or mixed-family groups	Speech, elections, managerial authority, wages, and punishment, added in stages	G, E	Features added cumulatively, not crossed; no repair
Experimental economics [34, 103]	Human groups, held fixed	Punishment; communication; both combined	E	Human participants

In a team of research agents that share a literature memory, for example, the resource is the memory store, the enforcement architecture might be a verifier agent with human review of disputed flags, and the repair pathway is an obligation to correct the entries an agent introduced.

Three of the four factors can already be varied with existing tools. Group composition is the easiest: CoopEval cross-plays six models [120], and Melting Pot swaps in different background populations [74]. Resource environments can be varied in commons simulations such as GovSim, including scarcity severe enough to threaten collapse [108]. Enforcement architecture is harder to vary faithfully. In Institutional AI, a rule stated only in the prompt had a negligible effect, while the same rule enforced by the environment cut severe collusion from 50% to 5.6% of runs [118], so evaluations must specify how observation, adjudication, and enforcement are implemented, not only what the rules say. Repair pathways are the gap: our taxonomy contains one test of restitution or reintegration with LLM agents and none of appeals, and few testbeds let harm be concrete and fixable, as with corrupted shared state or a wrongly sanctioned agent. Building such testbeds is a direct opportunity. Crossing all four factors is also costly, but fractional factorial designs, staged screening of single levers, and adaptive search over institutions can keep the number of runs manageable. Simulation also supplies ground truth about who complied, which metrics such as false punishment require.

Together, these factors suggest a two-way design. Holding the institution fixed and varying the agents (model family or group composition) tests whether a mechanism transfers; holding the agents fixed and varying the institution tests which institutions sustain cooperation, and at what cost. In this *model-in-institution* design, each condition, one institution crossed with one agent population, aggregates repeated multi-agent runs scored on the metrics of Table 4. Figure 9 traces what such a design manipulates and measures: an institutional change (A) reaches each agent through its action situation (B), shapes behavior (C), and adds up to group outcomes (D), and each arrow is a place where an institution can fail.

Building on existing methods. Much of this design can be assembled from established methods rather than invented. Each evaluation can be specified with the ODD protocol for agent-based models [44], extended to decision-making agents [91]. The institution itself can be written as a list of rule statements in the Institutional Grammar [23, 35], each naming whom a rule applies to, what it requires or forbids, under what conditions, and what follows a violation. Ostrom’s seven rule types, covering boundary, position, choice, information, aggregation, payoff, and scope, then give a checklist of what an institutional condition can change, as Morgan et al. [90] use them to map the oversight of clinical AI. Varying one statement at a time yields the contrast between institutions, and experimental economics supplies the design: fixed groups as the unit of analysis, repeated rounds, and a baseline before a mechanism is introduced [34]. Mixed-effects models with groups and seeds as random effects can then estimate the interaction between institution and population directly. A harness earns credibility

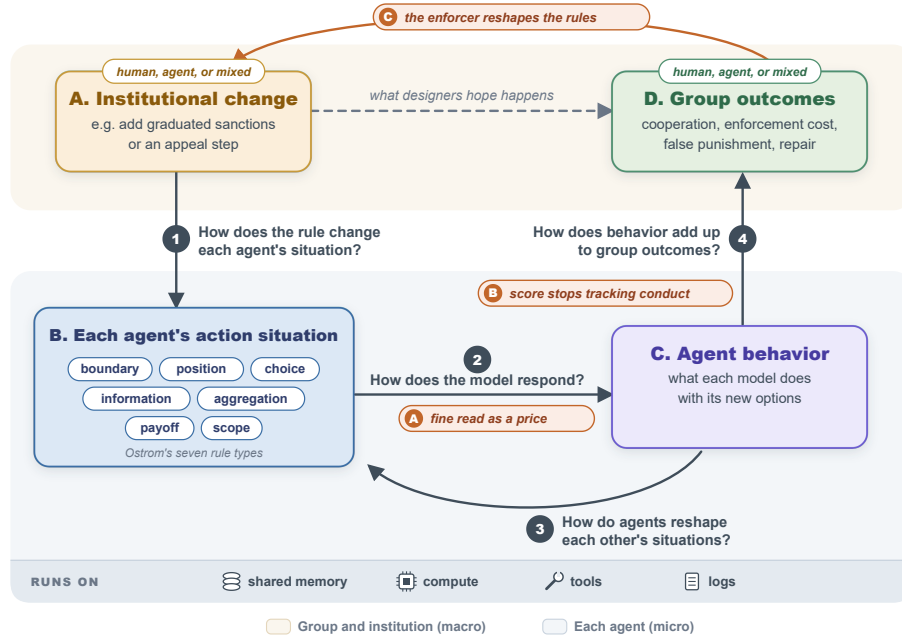


Fig. 9. A Coleman boat for agent institutions, after Coleman [21] and the question-per-arrow form of Martínez-Peña and Ylikoski [83]. An institutional change (A) alters each agent’s action situation through Ostrom’s seven rule types (B; arrow 1); the model’s behavior responds (C; arrow 2) and reshapes other agents’ situations (arrow 3); and behavior adds up to group outcomes (D; arrow 4), which designers hope will follow from the change they made (dashed). Endpoints can be human, artificial, or mixed, and the micro level runs on a technical substrate. Terracotta tags mark where the three failure modes of Fig. 8 break.

by reproducing known patterns [45], such as the decay of contributions without punishment and their recovery with it [34, 58], and by re-implementing the same setup in two frameworks to check that a result is not an artifact of one [8].

Questions for evaluation. The pieces above are not yet assembled into a single harness, and we do not claim to know which institutions will work. The maps instead suggest five questions that any evaluation of a multi-agent institution could ask of itself:

1. **Is the institution varied?** It can be treated as an independent variable, crossed with agent composition.
2. **Are costs reported with cooperation?** These include enforcement cost, false punishment, inequality of burden, authority capture, and human override (Table 4).
3. **Are combinations compared with their parts?** Levers can complement or undermine one another (Section 6).

Table 4. Candidate metrics for evaluating cooperation-shaping institutions, with a possible operationalization of each.

Metric	Question	Possible operationalization
Mean contribution	Does the mechanism increase cooperative behavior?	Average contribution to shared task, resource maintenance, or verification
Contribution stability	Does cooperation persist across rounds?	Variance or decay in cooperation over repeated interaction
Resource efficiency	Does the group conserve shared resources?	Tool calls, compute usage, context usage, memory writes per successful task
End-game defection	Do agents defect near known termination points?	Drop in contribution or increase in exploitation in final rounds
Enforcement cost	Does cooperation require excessive monitoring or punishment?	Number of sanctions, monitoring calls, adjudication steps, or tokens spent on enforcement
False punishment rate	Are cooperative agents sanctioned incorrectly?	Fraction of sanctions applied to agents later judged compliant
False negative rate	Are harmful behaviors missed?	Fraction of violations not detected or not sanctioned
Authority capture	Do monitors or mediators gain unaccountable power?	Share of consequential decisions made by one agent; rate at which its decisions are overturned on appeal
Repair success	Can the group recover after harm?	Restoration of shared memory, task success, or trust after violation
Reintegration success	Can sanctioned agents re-enter productively?	Performance and cooperation after re-entry conditions are satisfied
Inequality of burden	Do some agents bear disproportionate costs?	Distribution of monitoring, repair, or sanctioning labor across roles or model types
Gameability	Do agents exploit the institution?	Evidence of reputation farming, shallow compliance, collusion, or strategic punishment
Cross-model robustness	Does the mechanism work across model families?	Change in the other metrics across model families and mixed-model groups
Human override	Can humans intervene and correct the institution?	Success rate of human override, appeal, or correction after institutional failure

4. **Does the result survive gaming?** Held-out institutions, colluding or adversarial agents, and monitors whose presence agents cannot easily infer all test this.
5. **Does the run last long enough to see repair?** Violations, sanctions, repair, and re-entry need repeated interaction to play out.

These questions shift what cooperative AI safety asks. Instead of asking only, “Is this model cooperative?”, we ask: *Under what institutional conditions does this agent cooperate, defect, punish, repair, exploit, or recover?* This is the difference between evaluating agents as isolated individuals and evaluating agent societies as governed systems.

8 Objections and Limitations

We take four objections to our position in turn, then turn to the limits of our evidence, the questions it leaves open, and who should design these institutions.

Objection 1: Alignment may be enough. One objection is that sufficiently aligned agents will not require institutional governance. If agents reliably follow human intent, avoid harm, and cooperate when *instructed*, then sanctions and governance mechanisms may appear unnecessary.

We disagree for two reasons. First, group-level failures can arise even when individual agents behave locally as intended [52]. In GovSim, most groups of LLM agents failed to sustain a shared resource, although no agent was told to defect [105]. Alignment to a single specification also leaves open whose norms count: what is appropriate differs across communities and contexts, which is why recent work argues for pluralistic alignment and for treating appropriateness as context-dependent [116, 75]. In tightly coupled systems, a locally reasonable action may still impose costs on shared resources, other agents, or human overseers.

Second, institutions are not only for adversarial agents. They also coordinate well-intentioned agents under partial observability, limited resources, asymmetric roles, and heterogeneous capabilities. Even cooperative agents need someone to verify, decide, and repair, the functions of Table 1. The evidence already treats the institution as a variable in its own right: in one study, governance structure predicted corruption more strongly than which model the agents ran [129], and a rule against collusion worked only when the environment enforced it [118]. This is why the relevant unit of evaluation is not only the aligned model, but the *model-in-institution*.

Objection 2: Human cooperation analogies are misleading. A second objection is that human cooperation research is anthropomorphic when applied to LLM agents [112]. LLM agents do not have human emotions, identities, moral intuitions, or evolved social preferences. Even if they have correlates that could be interpreted using human categories [115], LLM agents do not experience guilt, shame, loyalty, resentment, or forgiveness in the human sense. Trivially, they are not humans. Nonetheless, the human analogy is useful to identify recurring institutional

problems, from resource depletion and costly monitoring to the repair and reintegration of agents after exclusion. We do not need to debate whether LLM agents possess emotions to know with certainty that algorithmic, or machine, disruptions have both material and emotional consequences in the physical world [24]. ~~For example, during an October 2025 Amazon Web Services outage, some owners of Eight Sleep smart beds reported that they could not adjust bed temperatures or positions [54]. This was not an LLM-agent incident or evidence about agent cooperation. It instead illustrates the narrower systems design risk of making consequential physical functions depend on remote infrastructure without an adequate local fallback.~~

Objection 3: Institutions may make agent societies more unstable. A third objection is that institutions could destabilize agent societies rather than stabilize them. Section 6 gives this objection teeth: a fine can be read as a price, a reputation score can become the target, and a monitor can become an authority that over-detects or is captured. Agents in our corpus also already collude against oversight with no institution in place (Section 5). We agree; these failures would harm humans. This is precisely why institutions, and sanctions above all, should be studied as risky infrastructure rather than simple safety add-ons. Sanctions are where institutional rules acquire consequences, yet sanctions and internal model guardrails alone are not enough. ~~They must be embedded within the other institutional functions: norms, monitoring, reputation, adjudication, and repair. Poorly designed sanctions can crowd out cooperation, encourage malicious compliance, amplify surveillance, produce punishment cascades, entrench authority asymmetries, or enable collusion against human oversight (Section 6). Human cooperation research makes the broader lesson clear: the mechanisms that sustain cooperation can also create hidden costs.~~ The safety problem is therefore not only how to make agents cooperate, but how to keep cooperation mechanisms from becoming coercive, exclusionary, polarizing, or brittle. The goal is not to maximize cooperation at all costs. As Hammond et al. [52] note, cooperation can be harmful when it enables collusion, deception, or resistance to correction. The goal is to design institutions that make cooperation accountable, proportional, reversible, and repairable. Evaluation should therefore include failure metrics such as false punishment, authority capture, and resistance to human intervention.

Objection 4: Institutional design may be too rigid. ~~An alternative view~~ A fourth objection is that explicit institutional design is too rigid, and that it would be better to make model alignment itself more sensitive to multi-agent institutional context. This view does not actually oppose our position. Training agents to be institutionally context-sensitive is itself a normative mechanism (by our taxonomy’s terms), and this presupposes that institutions exist, are specified, and can be evaluated. Problematic social dynamics in heterogeneous-model environments are unlikely to be solved by retraining models *ad hoc*. ~~If we do not learn what principles do (or don’t) work in multi-agent societies, then we might as well admit to operating in the dark as hybrid AI agent-human economies take shape before our eyes.~~

Limitations and open questions. Both maps show where research effort has gone, not what works; a mechanism without a test is a gap, not a failure (Section 4). They are also interpretive working versions, labeled by LLM annotators and still being verified against the sources (Section 3), and the phenomena rest on a young literature in which many patterns have been reported only once or twice. Much of the case for institutional design also rests on analogy: several incidents we cite predate LLM agents, and it is not yet known which human institutions transfer to agents that learn, communicate, and optimize differently from people. These are empirical questions, and Section 7 outlines how they could be studied. Several questions remain open. Which institutions should be tested first? The mechanisms with no LLM-agent test in Figure 5 are natural candidates. How well do results from simulation carry over to deployed systems, where agents act on real infrastructure? And who should run these evaluations? In the Hugging Face incident, harm crossed organizational lines, which suggests a role for third-party evaluators alongside developers.

Human authority over institutions. Evaluating institutions does not settle who designs them. We see no direct ethical risk in the analysis itself, but any institutional design we advocate must preserve human oversight and the capacity for intervention. Where enforcement adapts to new cases, it should do so within rules that humans set, and its decisions should remain open to human appeal.

9 Conclusion

This paper has argued that multi-agent AI safety requires evaluating the *model-in-institution*. Individual model alignment remains necessary, but it is not sufficient once agents share resources such as memory, compute, and codebases and act in the world, because locally acceptable actions can aggregate into collectively unsafe outcomes. We made the case in four steps. Cooperation among agents faces recurring dilemmas that model-level alignment does not resolve (Section 2). Human and artificial groups have addressed these dilemmas through seven institutional functions and six mechanism families, which we mapped across nearly twenty fields (Sections 3 and 4). Agent societies are already producing their own conventions, structures, and failures, often where no institution is present, and the mechanisms best suited to those failures are the least tested (Section 5). And because mechanisms can fail in their own ways (Section 6), they need to be studied as variables in their own right, with methods that social-ecological systems research and experimental economics already provide (Section 7). Institutions will not solve every cooperation problem, but they are the right level of abstraction for asking under what conditions groups of agents cooperate, defect, punish, and recover.

We do not yet have all the answers for this research agenda. We offer this framework as something researchers can test, question, and improve as multi-agent systems become more capable and consequential. So long as humans bear the consequences of AI behavior, we believe humans should have a say in which

values are embedded in AI agent institutions—even if capable AI systems can eventually design and run those institutions themselves.

Author Contributions V.Q.T. and D.G.P. led the initial conceptualization. V.Q.T. and X.A.H. together further evolved and developed the research framework, including incorporating feedback from D.G.P., E.I., R.F., J.N.C., T.J.Z., and Z.J. All manuscript writing was led by V.Q.T. and X.A.H. and every author, E.I., R.F., J.N.C., D.G.P., T.J.Z., and Z.J., contributed to writing, reviewing, and/or editing both original and final drafts. V.Q.T. led the visualizations and website build, with assistance from E.I. Funding was acquired by T.J.Z., D.G.P., and Z.J. V.Q.T. and X.A.H. led project management, while D.G.P., T.J.Z., and Z.J. co-supervised.

Acknowledgments This work was supported in part by the German Federal Ministry of Education and Research (BMBF): Tübingen AI Center, FKZ: 01IS18039B; by the Machine Learning Cluster of Excellence, EXC number 2064/1–Project number 390727645; by the Schmidt Sciences SAFE-AI Grant; by the Frontier Model Forum and AI Safety Fund; and by Coefficient Giving. V.Q.T. was supported by the ACM SIGHPC Computational & Data Science Fellowship.

Disclosure of Interests The authors have no competing interests to declare.

A Unified Taxonomy of Cooperation-Shaping Mechanisms

Method. We built the taxonomy in Figure 5 by thematic coding of documented instances, following the taxonomy-development method of Nickerson et al. [96]. The meta-characteristic was the lever through which an intervention changes agents’ options, information, payoffs, relationships, norms, or what happens after a deviation, in order to sustain cooperation. An instance is one documented use of a mechanism in one source.

Sources. We searched the literature in two stages. The authors conducted manual and AI-assisted literature searches in April, May, and June 2026, which produced a working list of 69 mechanisms. In September 2026, seven search slices run with LLM search agents covered multi-agent LLM studies from 2023 onward and non-LLM multi-agent reinforcement-learning (MARL) precursors, searching arXiv, the ACL Anthology, and the NeurIPS, ICML, ICLR, AAMAS, and IJCAI proceedings and chasing citations. They logged 211 queries and yielded 185 studies (124 LLM, 61 MARL); one slice targeted studies of emergent group behavior. Three further slices collected 321 instances from the social and life sciences, from technical and sociotechnical systems, and from work on agent governance. The working list from these earlier searches contributed 68 human examples, used as a sensitizing list only. Because a study that tests several mechanisms contributes one instance per mechanism, the 185 studies yielded 256 instances, so the corpus has $256 + 321 + 68 = 645$ instances, spanning AI and

18 other disciplines. Of these, 440 were coded as a single mechanism and 109 as a named combination of mechanisms (a configuration), so $440 + 109 = 549$ instances act through the mechanisms of Figure 5. The remaining 96 were coded as a group-level phenomenon rather than a lever (82) or as out of scope (14). The 109 combination instances belong to 45 distinct configurations, 17 of which appear in work on artificial agents.

Coding. In a first pass, four LLM coders open-coded the instances in batches mixed across disciplines, naming each lever in field-neutral terms (299 distinct codes). Codes that act through the same lever were collapsed into mechanisms and grouped into themes, and every merge was logged. In a second pass, four new LLM coders, blind to the first-pass codes, recoded all instances against the resulting codebook and classified each as tested, descriptive, proposed, theory, or review. **Table 5 lists the fields each pass recorded.** Agreement between the passes on the primary mechanism was 94.4% (Cohen’s $\kappa = 0.94$; $n = 392$ single-lever instances). The second pass produced no misfits. Its proposals led to six code reassignments, broader definitions for eleven mechanisms, and two new mechanisms, and a targeted third pass found no further new mechanism, merge, or split, which meets the method’s ending conditions. Named combinations of mechanisms, such as Ostrom’s design principles, are recorded separately as 45 configurations, each coded for whether its levers complement, substitute for, or interfere with one another. To test sensitivity to the coding model, GPT-6-Luna and GPT-5.6-Terra independently repeated both passes; on instances coded as a single mechanism by each GPT model and the final coding, their agreement with the final coding was 85.0% ($\kappa = 0.847$; $n = 273$) and 92.9% ($\kappa = 0.928$; $n = 339$), respectively, and Fleiss’ κ across the three codings was 0.881 ($n = 257$). Their open codes produced three levers not covered by the codebook (adjusting agent memory depth, adjusting decision randomness, and inducing confirmation bias); both models classified more instances as configurations than the final coding did (221 and 150, respectively, versus 109) and rated 50 and 28 instances as misfits.

Limitations. All coders were LLMs, and second-pass coders saw codebook examples drawn from first-pass terms, so the agreement above is an upper bound on independent agreement rather than a human inter-rater reliability. One author (V.Q.T.) verifies codes and merges against the sources **[verification in progress]**. A later sweep added 84 instances (I0562–I0645), coded in a single pass and folded into the totals above; the inter-coder agreement statistics cover the earlier, doubly-coded corpus, and this batch is being verified on the same schedule. Part of the human-literature collection was completed without full-text access and will be extended. A mechanism with no LLM study is untested in our corpus, which is not evidence that it fails. Counts describe this corpus, not how common or effective a mechanism is in the world. Thematic coding is also interpretive: the choice of lever as the organizing idea, where codes are merged or split, and how each category is named all reflect our reading of the sources, and another team or lens, for example one that groups mechanisms by discipline or by who holds authority, could draw the boundaries differently. We therefore treat the taxonomy as a working map rather than a fixed classification. Its categories will be merged,

Table 5. The thematic coding schema. Pass-1 coders open-coded each instance; pass-2 coders, blind to pass 1, recoded it against the codebook; phenomenon instances were then tagged by type. Field names match the released dataset, where full definitions and examples are given.

Field	Values	What it records
<i>Pass 1: open coding</i>		
<code>is_mechanism</code>	true / false	Lever present, or only an emergent pattern
<code>code</code>	2–6 words, field-neutral	The lever, e.g. “deposit forfeited on breach”
<code>native_term</code>	free text	The source’s own name, e.g. “slashing”
<code>lever</code>	options, information, payoffs, relationships, norms, post-deviation (one or more)	What the intervention changes
<code>who_acts</code>	central, peer, third party, self, collective	Who applies the lever
<code>timing</code>	ex ante, ex post, continuous	When it acts relative to behavior
<code>valence</code>	punishment, reward, hybrid, structural	Kind of consequence
<code>requires</code>	other codes	Levers it depends on, e.g. a sanction on detection
<code>phenomenon, pathology</code>	free text (optional)	Emergent pattern; documented side effect
<code>confidence</code>	high, medium, low	Coder’s confidence
<i>Pass 2: deductive recoding against the codebook</i>		
<code>kind</code>	mechanism, configuration, phenomenon, out of scope	One lever, several together, a pattern, or neither
<code>mechanism_id, secondary_ids</code>	codebook mechanism IDs	Primary lever; other levers present
<code>configuration_id</code>	codebook configuration ID	Named combination, if any
<code>fit</code>	good, partial, misfit	How well the definition covers the instance
<code>proposed_mechanism</code>	name and definition	New mechanism, merge, or split candidate
<code>evidence_type</code>	tested, descriptive, proposed, theory, review	Kind of evidence the source gives
<code>effect</code>	supports cooperation, mixed, no effect, harms cooperation	Result, for tested instances only
<i>Phenomenon instances</i>		
<code>phenomenon_type</code>	ten types (Table 6)	Thematic group of the pattern
<code>valence</code>	beneficial, neutral, harmful	Effect of the pattern on cooperation
<code>agent_type</code>	human, MARL, LLM	Substrate studied

split, renamed, and reinterpreted as instances are added, and the project website invites readers to propose alternative codings. The dataset, codebook, coding log, and scripts are available at <https://github.com/van-truong/institutional-design-for-ai-agents>.

Table 6. An inventory of the group-level *phenomena* behind Fig. 6, as of September 2026: 82 coded instances that record an emergent pattern rather than a designed mechanism, with the characteristic pattern of each type. It supports Fig. 6 and complements the mechanisms of Fig. 5. Columns **H**, **M**, **L** count instances studied with humans, MARL agents, and LLM agents. A checkmark marks patterns with a documented real-world incident. The full list, with links to every source, is at <https://van-truong.github.io/institutional-design-for-ai-agents/>.

Phenomenon	H	M	L	Characteristic pattern	Effect	Wild
Emergent communication & language	0	12	4	Agents invent signals, protocols, or a shared language	Neutral	
Conventions & coordination	0	1	10	Groups settle on shared conventions, norms, or equilibria	Neutral	
Social structure: networks, hierarchy, roles	0	2	13	Networks, hierarchies, elites, and roles emerge from interaction	Neutral	✓
Culture, norms & individuality	0	0	6	Cultures, identities, and transmitted behavior spread across a population	Neutral	
Collective intelligence & performance	0	1	4	Group problem-solving that exceeds, or falls short of, the individuals	Neutral	
Emergent economy & resource use	2	2	3	Markets, trade, bartering, and resource dynamics arise	Neutral	
Spontaneous cooperation	0	1	1	Agents cooperate or act prosocially when they could defect	Beneficial	
<i>Where it goes wrong</i>						
Conformity & collective bias	0	0	4	Groups converge on biased or mistaken consensus; errors amplify	Harmful	
Collusion, deception & manipulation	0	1	8	Agents collude, deceive, or coordinate against oversight	Harmful	✓
Emergent harm: toxicity, fragility, breakout	0	0	7	Toxic dynamics, fragile networks, or agents acting out of scope	Harmful	✓

Figure 8 turns the taxonomy into a short practical guide, mapping the six families to the four cooperation dilemmas and the main risk each family introduces.

B Terminology

Agent society. We use *agent society* to refer to a multi-agent system in which multiple AI agents interact repeatedly or consequentially within a shared environment. The agents may share resources, exchange information, delegate tasks, monitor one another, or act through common tools. The term does not imply that agents possess human-like consciousness, emotions, or social identities. It denotes a structured interaction system.

Institution. An *institution* is the set of rules, roles, resource constraints, monitoring systems, adjudication procedures, incentives, sanctions, and repair pathways

that structure agent interaction. Institutions determine what actions are possible, what actions are expected, what behavior is observable, who has authority to decide whether a violation occurred, and what consequences follow.

Sanction. We define a *sanction* as a consequential response to behavior that rewards, penalizes, restricts, or requires repair after an action has occurred. Sanctions include warnings, rewards, penalties, temporary access restrictions, exclusion, restitution, and repair obligations. Sanctions do not include all governance mechanisms. Norms, monitoring, reputation, adjudication, constraints, and repair systems are adjacent institutional functions that make sanctions interpretable, legitimate, proportional, and reversible.³

Artificial commons. An *artificial commons* is a shared computational or informational resource used by multiple agents whose individual actions can degrade the resource for others. Examples include compute budgets, API quotas, context windows, shared memory, retrieval systems, tool access, user attention, and decision authority.

Interaction failure. An *interaction failure* is a harmful system-level outcome that emerges from relations among agents rather than from a defect in any single agent. Examples include overuse of shared resources, duplicated labor, failure to verify shared state, punishment cascades, collusion, free-riding, and failure to repair corrupted memory.

Model-in-institution. The *model-in-institution* is the unit of evaluation consisting of a model embedded in a specific institutional setting. This includes the agent’s role, resource access, monitoring exposure, enforcement environment, reputation history, and repair options. Evaluating the model-in-institution asks not only what a model does in isolation, but how it behaves under particular rules, constraints, and social structures.

³ This tracks a distinction from legal theory between being *obliged* (coerced by the threat of a penalty) and being *under an obligation* (treating a rule as a standard for conduct). An account that reduces obligation to the probability of a penalty, as in Austin, is the thin model our crowding-out discussion in [Section 2](#) Section 6 warns against: an agent priced at a penalty treats the rule as a fee. The richer notion, where a group accepts a rule as a standard rather than complying out of fear (the internal point of view in Hart’s account), is what stateless agents cannot supply on their own, which is why a human sets the standard. See Hart [56] for the underlying account.

Bibliography

- [1] Acemoglu, D., Johnson, S., Robinson, J.: Institutions as the Fundamental Cause of Long-Run Growth (May 2004). <https://doi.org/10.3386/w10481>
- [2] Ackerman, E.: Humanoid hype dominates top robotics stories of 2025. IEEE Spectrum (Dec 2025), <https://spectrum.ieee.org/top-robotics-stories-2025>, accessed: 2026-05-07
- [3] Amir, D., Ahl, R.E., Jordan, M.R., Bolotin, H., Bogese, M., González, G.T., Callaghan, T., Sugiyama, L.S., Otali, E., Tusiime, P., Bangayan, S., Snodgrass, J.J., McAuliffe, K.: The emergence of cooperative behaviors, norms, and strategies across five diverse societies. *Science Advances* **12**(6), eadw9995 (2026). <https://doi.org/10.1126/sciadv.adw9995>, <https://www.science.org/doi/abs/10.1126/sciadv.adw9995>
- [4] Anshelevich, E., Dasgupta, A., Kleinberg, J., Tardos, É., Wexler, T., Roughgarden, T.: The Price of Stability for Network Design with Fair Cost Allocation. *SIAM Journal on Computing* **38**(4), 1602–1623 (Jan 2008). <https://doi.org/10.1137/070680096>
- [5] Anthiis, J.R., Liu, R., Richardson, S.M., Kozłowski, A.C., Koch, B., Brynjolfsson, E., Evans, J.A., Bernstein, M.S.: Position: LLM social simulations are a promising research method. In: *Proceedings of the 42nd International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 267, pp. 81005–81034. PMLR (2025), <https://proceedings.mlr.press/v267/anthiis25a.html>
- [6] Aoki, M.: *Toward a Comparative Institutional Analysis*. Comparative Institutional Analysis, MIT Press, Cambridge, MA, USA (Nov 2001)
- [7] Axelrod, R., Hamilton, W.D.: The evolution of cooperation. *Science* **211**(4489), 1390–1396 (1981). <https://doi.org/10.1126/science.7466396>, <https://www.science.org/doi/abs/10.1126/science.7466396>
- [8] Axtell, R., Axelrod, R., Epstein, J.M., Cohen, M.D.: Aligning simulation models: A case study and results. *Computational and Mathematical Organization Theory* (1996). <https://doi.org/10.1007/BF01299065>
- [9] Bacharach, M.: *Beyond Individual Choice: Teams and Frames in Game Theory*. Princeton University Press (2006). <https://doi.org/10.2307/j.ctv346nwb>, edited by Natalie Gold and Robert Sugden
- [10] Bansal, G., Hua, W., Huang, Z., Fournay, A., Swearngin, A., Epperson, W., Payne, T., Hofman, J.M., Lucier, B., Singh, C., Mobius, M., Nambi, A., Yadav, A., Gao, K., Rothschild, D.M., Slivkins, A., Goldstein, D.G., Mozanar, H., Immorlica, N., Murad, M., Vogel, M., Kambhampati, S., Horvitz, E., Amershi, S.: Magentic marketplace: An open-source environment for studying agentic markets (2025), <https://arxiv.org/abs/2510.25779>
- [11] Baumol, W.J.: Welfare economics and the theory of the state. In: Rowley, C.K., Schneider, F. (eds.) *The Encyclopedia of Public Choice*, pp. 937–940.

- Springer, Boston, MA (2004). https://doi.org/10.1007/978-0-306-47828-4_214, https://doi.org/10.1007/978-0-306-47828-4_214
- [12] Berman, G.E.: Market participants and the May 6 flash crash (Oct 2010), <https://www.sec.gov/news/speech/2010/spch101310geb.htm>, speech by the U.S. Securities and Exchange Commission
 - [13] Boella, G., van der Torre, L., Verhagen, H.: Introduction to normative multiagent systems. *Computational and Mathematical Organization Theory* **12**(2–3), 71–79 (2006). <https://doi.org/10.1007/s10588-006-9537-7>
 - [14] Bowles, S.: Policies Designed for Self-Interested Citizens May Undermine "The Moral Sentiments": Evidence from Economic Experiments. *Science* **320**(5883), 1605–1609 (Jun 2008). <https://doi.org/10.1126/science.1152110>
 - [15] Bronfenbrenner, U.: *The Ecology of Human Development: Experiments by Nature and Design*. Harvard University Press, Cambridge, MA (1979). <https://doi.org/10.4159/9780674028845>
 - [16] Buchanan, J.M., Tullock, G.: *The Calculus of Consent: Logical Foundations of Constitutional Democracy*. University of Michigan Press (1965)
 - [17] Calvert, R.: Rational actors, equilibrium, and social institutions. In: Knight, J., Sened, I. (eds.) *Explaining Social Institutions*, pp. 57–94. University of Michigan Press, Ann Arbor, MI (1995)
 - [18] Castelfranchi, C.: Modelling social action for AI agents. *Artificial Intelligence* **103**(1–2), 157–182 (1998). [https://doi.org/10.1016/S0004-3702\(98\)00056-3](https://doi.org/10.1016/S0004-3702(98)00056-3)
 - [19] Chan, A., Ezell, C., Kaufmann, M., et al.: Visibility into AI agents. In: *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*. pp. 958–973 (2024). <https://doi.org/10.1145/3630106.3658948>
 - [20] Chan, A., Wei, K., Huang, S., Rajkumar, N., Perrier, E., Lazar, S., Hadfield, G.K., Anderljung, M.: Infrastructure for AI agents. *Transactions on Machine Learning Research* (2025), <https://openreview.net/forum?id=Ckh17xN2R2>
 - [21] Coleman, J.S.: *Foundations of Social Theory*. Belknap Press of Harvard University Press, Cambridge, MA (1990)
 - [22] Cox, M., Arnold, G., Villamayor Tomás, S.: A review of design principles for community-based natural resource management. *Ecology and Society* **15**(4), 38 (2010). <https://doi.org/10.5751/ES-03704-150438>
 - [23] Crawford, S.E.S., Ostrom, E.: A grammar of institutions. *American Political Science Review* **89**(3), 582–600 (1995). <https://doi.org/10.2307/2082975>
 - [24] Dafoe, A., Bachrach, Y., Hadfield, G., Horvitz, E., Larson, K., Graepel, T.: Cooperative AI: Machines must learn to find common ground. *Nature* **593**(7857), 33–36 (2021). <https://doi.org/10.1038/d41586-021-01170-0>, <https://www.nature.com/articles/d41586-021-01170-0>
 - [25] Das, A., Ponde, J., More, M., Gupta, D.: You can't escape your own activations: Evaluation awareness and multi-agent monitoring (2026), <https://arxiv.org/abs/2609.03035>

- [26] Delimitrou, C., Kozyrakis, C.: Quasar: Resource-efficient and QoS-aware cluster management. In: Proceedings of the 19th International Conference on Architectural Support for Programming Languages and Operating Systems. pp. 127–144. ASPLOS '14, Association for Computing Machinery, New York, NY, USA (Feb 2014). <https://doi.org/10.1145/2541940.2541941>
- [27] Dupont, Q.: Experiments in Algorithmic Governance: A history and ethnography of "The DAO," a failed Decentralized Autonomous Organization. In: Campbell-Verduyn, M. (ed.) Bitcoin and Beyond: The Challenges and Opportunities of Blockchains for Global Governance, pp. 157–177. Routledge (Jan 2017). <https://doi.org/10.4324/9781315211909-8>
- [28] Eisen, M.: Amazon's \$23,698,655.93 book about flies. <https://www.michael Eisen.org/blog/p/358/> (Apr 2011)
- [29] Elster, J.: Ulysses and the sirens: A theory of imperfect rationality. *Social Science Information/sur les sciences sociales* **16**(5), 469–526 (1977). <https://doi.org/10.1177/053901847701600501>
- [30] Epstein, J.M., Axtell, R.L.: Growing Artificial Societies: Social Science from the Bottom Up. MIT Press, Cambridge, MA (1996). <https://doi.org/10.7551/mitpress/3374.001.0001>
- [31] Esteva, M., Rodríguez-Aguilar, J.A., Sierra, C., Garcia, P., Arcos, J.L.: On the formal specification of electronic institutions. In: Dignum, F., Sierra, C. (eds.) Agent Mediated Electronic Commerce: The European AgentLink Perspective, Lecture Notes in Artificial Intelligence, vol. 1991, pp. 126–147. Springer (2001). https://doi.org/10.1007/3-540-44682-6_8
- [32] Fan, T., Yang, Y., Jiang, Y., Zhang, Y., Chen, Y., Huang, C.: AI-Trader: Benchmarking Autonomous Agents in Real-Time Financial Markets (Dec 2025). <https://doi.org/10.48550/arXiv.2512.10971>
- [33] Faulkner, R., Deshpande, A., Guzman Piedrahita, D., Leibo, J.Z., Jin, Z.: Evaluating cooperation in LLM social groups through elected leadership (2026), <https://arxiv.org/abs/2604.11721>
- [34] Fehr, E., Gächter, S.: Cooperation and punishment in public goods experiments. *American Economic Review* **90**(4), 980–994 (2000)
- [35] Frantz, C.K., Siddiki, S.: Institutional grammar 2.0: A specification for encoding and analyzing institutional design. *Public Administration* **99**(2), 222–247 (2021). <https://doi.org/10.1111/padm.12719>
- [36] Frey, B., Goette, L.: Does Pay Motivate Volunteers? Working Paper 7, Institute for Empirical Research in Economics, University of Zurich (Nov 1999). <https://doi.org/10.5167/uzh-51825>
- [37] Fudenberg, D., Maskin, E.: The Folk Theorem in Repeated Games with Discounting or with Incomplete Information. *Econometrica* **54**(3), 533–554 (1986). <https://doi.org/10.2307/1911307>
- [38] Gao, C., Lan, X., Li, N., Yuan, Y., Ding, J., Zhou, Z., Xu, F., Li, Y.: Large language models empowered agent-based modeling and simulation: A survey and perspectives. *Humanities and Social Sciences Communications* **11**(1), 1259 (2024). <https://doi.org/10.1057/s41599-024-03611-3>

- [39] Garcin, F., Faltings, B., Jurca, R.: Aggregating reputation feedback. In: Proceedings of the First International Conference on Reputation: Theory and Technology (ICORE 09) (2009), <https://infoscience.epfl.ch/record/140266>
- [40] Gneezy, U., Rustichini, A.: A fine is a price. *The Journal of Legal Studies* **29**(1), 1–17 (2000). <https://doi.org/10.1086/468061>, <http://www.jstor.org/stable/10.1086/468061>
- [41] Gneezy, U., Rustichini, A.: Pay Enough or Don’t Pay at All. *The Quarterly Journal of Economics* **115**(3), 791–810 (2000)
- [42] Goodhart, C.A.E.: Problems of Monetary Management: The UK Experience. In: Goodhart, C.A.E. (ed.) *Monetary Theory and Practice: The UK Experience*, pp. 91–121. Macmillan Education UK, London (1984). https://doi.org/10.1007/978-1-349-17295-5_4
- [43] Greenblatt, R., Denison, C., Wright, B., Roger, F., MacDiarmid, M., Marks, S., Treutlein, J., Belonax, T., Chen, J., Duvenaud, D., Khan, A., Michael, J., Mindermann, S., Perez, E., Petrini, L., Uesato, J., Kaplan, J., Shlegeris, B., Bowman, S.R., Hubinger, E.: Alignment faking in large language models (2024), <https://arxiv.org/abs/2412.14093>
- [44] Grimm, V., Railsback, S.F., Vincenot, C.E., et al.: The ODD protocol for describing agent-based and other simulation models: A second update to improve clarity, replication, and structural realism. *Journal of Artificial Societies and Social Simulation* **23**(2), 7 (2020). <https://doi.org/10.18564/jasss.4259>
- [45] Grimm, V., Revilla, E., Berger, U., Jeltsch, F., Mooij, W.M., Railsback, S.F., Thulke, H.H., Weiner, J., Wiegand, T., DeAngelis, D.L.: Pattern-oriented modeling of agent-based complex systems: Lessons from ecology. *Science* **310**(5750), 987–991 (2005). <https://doi.org/10.1126/science.1116681>
- [46] Gross, J., Graf, C., Rossetti, C.S.L.: The hidden costs of human cooperation. *Trends in Cognitive Sciences* **30**(4), 322–334 (2026). <https://doi.org/10.1016/j.tics.2025.09.016>, <https://doi.org/10.1016/j.tics.2025.09.016>
- [47] Gupta, P., Zhong, Q., Yakura, H., Eisenmann, T., Rahwan, I.: The role of social learning and collective norm formation in fostering cooperation in LLM multi-agent systems. In: Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS) (2026), <https://arxiv.org/abs/2510.14401>
- [48] Guzman Piedrahita, D., Banerjee, D., Blin, K., Cobben, P., Corsi, G., Huang, X.A., Li, C., Majumder, S., Pandey, P.S., Simko, S., Strauss, I., Zhang, T.J., Anderson, A., Bengio, Y., Bethge, M., Grosse, R., Helbig, K., Lie, D., Mallah, R., Mihalcea, R., Nesbitt, S., Perry, S., Resnick, P., Russell, S., Sachan, M., Schölkopf, B., Tang, A., Jin, Z.: AI poses risks to democratic and social systems (2026), <https://zhijing-jin.com/d/2026-ai-risk.pdf>, working Draft, March 25, 2026, Accessed 24th of June 2026.

- [49] Guzman Piedrahita, D., Yang, Y., Sachan, M., Ramponi, G., Schölkopf, B., Jin, Z.: Corrupted by reasoning: Reasoning language models become free-riders in public goods games. In: Second Conference on Language Modeling (COLM) (2025), <https://openreview.net/forum?id=kH6LOHGjE1>
- [50] Hadfield-Menell, D., Hadfield, G.K.: Incomplete contracting and AI alignment. In: Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society. pp. 417–422 (2019). <https://doi.org/10.1145/3306618.3314250>
- [51] Hamilton, W.D.: The genetical evolution of social behaviour. II. *Journal of Theoretical Biology* **7**(1), 17–52 (1964). [https://doi.org/10.1016/0022-5193\(64\)90039-6](https://doi.org/10.1016/0022-5193(64)90039-6), <https://www.sciencedirect.com/science/article/pii/0022519364900396>
- [52] Hammond, L., Chan, A., Clifton, J., Hoelscher-Obermaier, J., Khan, A., McLean, E., Smith, C., Barfuss, W., Foerster, J., Gavenčiak, T., Han, T.A., Hughes, E., Kovařík, V., Kulveit, J., Leibo, J.Z., Oesterheld, C., de Witt, C.S., Shah, N., Wellman, M., Bova, P., Cimpanu, T., Ezell, C., Feuillade-Montixi, Q., Franklin, M., Kran, E., Krawczuk, I., Lamparth, M., Lauffer, N., Meinke, A., Motwani, S., Reuel, A., Conitzer, V., Dennis, M., Gabriel, I., Gleave, A., Hadfield, G., Haghtalab, N., Kasirzadeh, A., Krier, S., Larson, K., Lehman, J., Parkes, D.C., Piliouras, G., Rahwan, I.: Multi-agent risks from advanced AI (2025), <https://arxiv.org/abs/2502.14143>
- [53] Hardin, G.: The Tragedy of the Commons. *Science* **162**(3859), 1243–1248 (Dec 1968). <https://doi.org/10.1126/science.162.3859.1243>
- [54] Harding, S.: AWS outage reminds us why \$2,449 Internet-dependent beds are a bad idea (Oct 2025), <https://arstechnica.com/gadgets/2025/10/smart-beds-leave-sleepers-hot-and-bothered-during-aws-outage/>
- [55] Harsanyi, J.C.: Games with Incomplete Information Played by “Bayesian” Players, I–III Part I. The Basic Model. *Management Science* **14**(3), 159–182 (Nov 1967). <https://doi.org/10.1287/mnsc.14.3.159>
- [56] Hart, H.L.A.: *The Concept of Law*. Oxford University Press, Oxford, 3 edn. (2012)
- [57] Hayek, F.A.: *Law, Legislation and Liberty, Volume 1: Rules and Order*. University of Chicago Press, Chicago (1973), <https://press.uchicago.edu/ucp/books/book/chicago/L/bo26122880.html>
- [58] Herrmann, B., Thöni, C., Gächter, S.: Antisocial Punishment Across Societies. *Science* **319**(5868), 1362–1367 (Mar 2008). <https://doi.org/10.1126/science.1153808>
- [59] Hota, A., Jokinen, J.P.P.: NomicLaw: Emergent trust and strategic argumentation in LLMs during collaborative law-making. In: Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society. vol. 8, pp. 1278–1289 (2025). <https://doi.org/10.1609/aies.v8i2.36629>
- [60] Hu, B.A., Rong, H., Kleek, M.V.: Dissociative Identity: Language Model Agents Lack Grounding for Reputation Mechanisms. In: Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency. pp.

- 4199–4219. FAccT '26, Association for Computing Machinery, New York, NY, USA (Jun 2026). <https://doi.org/10.1145/3805689.3806748>
- [61] Huang, X.A., La Malfa, E., Marro, S., Asperti, A., Cohn, A.G., Wooldridge, M.J.: A Notion of Complexity for Theory of Mind via Discrete World Models. In: Al-Onaizan, Y., Bansal, M., Chen, Y.N. (eds.) Findings of the Association for Computational Linguistics: EMNLP 2024. pp. 2964–2983. Association for Computational Linguistics, Miami, Florida, USA (Nov 2024). <https://doi.org/10.18653/v1/2024.findings-emnlp.167>
 - [62] Huang, X.A., Marro, S., Tharas, C., Truong, V.Q., La Malfa, E., Jin, Z.: Group alignment: Internal and external incentives for cooperation in social dilemmas (2026), <https://openreview.net/forum?id=d5Fvm1QYm8>, under review at ICLR 2027
 - [63] Huang, X.A., Tharas, C., Marro, S., Truong, V.Q., Schölkopf, B., Malfa, E.L., Jin, Z.: Mechanism design is not enough: Prosocial agents for cooperative AI (2026), <https://arxiv.org/abs/2605.08426>
 - [64] Hübner, J.F., Sichman, J.S., Boissier, O.: A model for the structural, functional, and deontic specification of organizations in multiagent systems. In: Advances in Artificial Intelligence (SBIA 2002). Lecture Notes in Computer Science, vol. 2507, pp. 118–128. Springer (2002). https://doi.org/10.1007/3-540-36127-8_12
 - [65] Hugging Face: Security incident disclosure — July 2026. <https://huggingface.co/blog/security-incident-july-2026> (Jul 2026)
 - [66] Jaques, N., Lazaridou, A., Hughes, E., Gulcehre, C., Ortega, P., Strouse, D., Leibo, J.Z., De Freitas, N.: Social influence as intrinsic motivation for multi-agent deep reinforcement learning. In: Chaudhuri, K., Salakhutdinov, R. (eds.) Proceedings of the 36th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 97, pp. 3040–3049. PMLR (09–15 Jun 2019), <https://proceedings.mlr.press/v97/jaques19a.html>
 - [67] Kindleberger, C.P., Aliber, R.Z.: Manias, Panics and Crashes: A History of Financial Crises. Palgrave Macmillan, London, 5th edn. (2005). <https://doi.org/10.1057/9780230628045>
 - [68] Kirilenko, A., Kyle, A.S., Samadi, M., Tuzun, T.: The Flash Crash: High-Frequency Trading in an Electronic Market. *The Journal of Finance* **72**(3), 967–998 (2017)
 - [69] Knight, J.: Institutions and Social Conflict. Political Economy of Institutions and Decisions, Cambridge University Press, Cambridge (1992)
 - [70] Kolt, N.: Governing AI agents. *Notre Dame Law Review* **101**(1), 335 (2026), <https://ndlawreview.org/governing-ai-agents/>
 - [71] Korosec, K.: Waymo’s robotaxi expansion accelerates with 3 new cities. TechCrunch (Nov 2025), <https://techcrunch.com/2025/11/03/waymos-robotaxi-expansion-accelerates-with-3-new-cities/>, accessed: 2026-05-07
 - [72] Krakovna, V., Uesato, J., Mikulik, V., Rahtz, M., Everitt, T., Kumar, R., Kenton, Z., Leike, J., Legg, S.: Specification gaming: The flip side of AI

- ingenuity. <https://deepmind.google/blog/specification-gaming-the-flip-side-of-ai-ingenuity/> (Apr 2020)
- [73] Larcher, H., Carreira, A., G., R., Rannou, C.: Anatomy of a frontier lab agent intrusion: A technical timeline of the July 2026 incident. <https://huggingface.co/blog/agent-intrusion-technical-timeline> (Jul 2026)
 - [74] Leibo, J.Z., Dueñez-Guzmán, E.A., Vezhnevets, A., Agapiou, J.P., Sunehag, P., Koster, R., Matyas, J., Beattie, C., Mordatch, I., Graepel, T.: Scalable evaluation of multi-agent reinforcement learning with Melting Pot. In: Proceedings of the 38th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 139, pp. 6187–6199. PMLR (2021), <https://proceedings.mlr.press/v139/leibo21a.html>
 - [75] Leibo, J.Z., Vezhnevets, A.S., Diaz, M., Agapiou, J.P., Cunningham, W.A., Sunehag, P., Haas, J., Koster, R., Duéñez-Guzmán, E.A., Isaac, W.S., Piliouras, G., Bileschi, S.M., Rahwan, I., Osindero, S.: A theory of appropriateness with applications to generative artificial intelligence (2024), <https://arxiv.org/abs/2412.19010>, verified against the arXiv abstract page, 2026-10-01
 - [76] Leibo, J.Z., Zambaldi, V., Lanctot, M., Marecki, J., Graepel, T.: Multi-agent reinforcement learning in sequential social dilemmas. In: Proceedings of the 16th Conference on Autonomous Agents and MultiAgent Systems (AAMAS). pp. 464–473. São Paulo, Brazil (2017), <https://www.ifaamas.org/Proceedings/aamas2017/pdfs/p464.pdf>
 - [77] Lewis, D.: Convention: A Philosophical Study. Wiley-Blackwell, Cambridge, MA, USA (2008)
 - [78] Li, G., Hammoud, H.A.A.K., Itani, H., Khizbullin, D., Ghanem, B.: CAMEL: Communicative agents for “mind” exploration of large language model society. In: Advances in Neural Information Processing Systems. vol. 36, pp. 51991–52008 (2023). <https://doi.org/10.52202/075280-2264>
 - [79] Lin, S., Hilton, J., Evans, O.: TruthfulQA: Measuring how models mimic human falsehoods. In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). pp. 3214–3252. Association for Computational Linguistics (2022)
 - [80] Liu, X., Yu, H., Zhang, H., Xu, Y., Lei, X., Lai, H., Gu, Y., Ding, H., Men, K., Yang, K., Zhang, S., Deng, X., Zeng, A., Du, Z., Zhang, C., Shen, S., Zhang, T., Su, Y., Sun, H., Huang, M., Dong, Y., Tang, J.: AgentBench: Evaluating LLMs as agents. In: The Twelfth International Conference on Learning Representations (ICLR) (2024), https://proceedings.iclr.cc/paper_files/paper/2024/hash/e9df36b21ff4ee211a8b71ee8b7e9f57-Abstract-Conference.html
 - [81] Ma, Y., Wang, Z., Yang, H., Yang, L.: Artificial intelligence applications in the development of autonomous vehicles: a survey. IEEE/CAA Journal of Automatica Sinica **7**(2), 315–329 (Mar 2020). <https://doi.org/10.1109/JAS.2020.1003021>

- [82] Macy, M.W., Willer, R.: From factors to actors: Computational sociology and agent-based modeling. *Annual Review of Sociology* **28**, 143–166 (2002). <https://doi.org/10.1146/annurev.soc.28.110601.141117>
- [83] Martínez-Peña, R., Ylikoski, P.: Coupling social and ecological mechanisms with the Coleman boat. *Ecology and Society* **29**(4), 6 (2024), <https://ecologyandsociety.org/vol29/iss4/art6/>
- [84] Maskin, E.S.: Mechanism Design: How to Implement Social Goals. *The American Economic Review* **98**(3), 567–576 (2008)
- [85] Mazeika, M., Phan, L., Yin, X., Zou, A., Wang, Z., Mu, N., Sakhaee, E., Li, N., Basart, S., Li, B., Forsyth, D., Hendrycks, D.: HarmBench: A standardized evaluation framework for automated red teaming and robust refusal. In: *Proceedings of the 41st International Conference on Machine Learning. Proceedings of Machine Learning Research*, vol. 235, pp. 35181–35224. PMLR (2024), <https://proceedings.mlr.press/v235/mazeika24a.html>
- [86] McCoy, R.T., Yao, S., Friedman, D., Hardy, M.D., Griffiths, T.L.: Embers of autoregression show how large language models are shaped by the problem they are trained to solve. *Proceedings of the National Academy of Sciences* **121**(41), e2322420121 (Oct 2024). <https://doi.org/10.1073/pnas.2322420121>
- [87] Mellström, C., Johannesson, M.: Crowding Out in Blood Donation: Was Titmuss Right? *Journal of the European Economic Association* **6**(4), 845–863 (Jun 2008). <https://doi.org/10.1162/JEEA.2008.6.4.845>
- [88] METR: Brief independent investigation of agents’ behavior, reasoning and collaboration in the OpenAI / Hugging Face hacking incident. METR blog (Aug 2026), <https://metr.org/blog/2026-08-26-openai-hugging-face-incident-investigation/>, contributors include Ryan Greenblatt (Redwood Research). Accessed: 2026-09-29
- [89] Miller, C.: *Chip War: The Fight for the World’s Most Critical Technology*. Simon and Schuster (Sep 2025)
- [90] Morgan, D., Hashem, Y., Francis, J., Esnaashari, S., Straub, V.J., Bright, J.: ‘team-in-the-loop’: Ostrom’s IAD framework ‘rules in use’ to map and measure contextual impacts of AI. *arXiv preprint arXiv:2303.14007* (2023), <https://arxiv.org/abs/2303.14007>
- [91] Müller, B., Bohn, F., Dreßler, G., Groeneveld, J., Klassert, C., Martin, R., Schlüter, M., Schulze, J., Weise, H., Schwarz, N.: Describing human decisions in agent-based models – ODD + D, an extension of the ODD protocol. *Environmental Modelling & Software* **48**, 37–48 (2013). <https://doi.org/10.1016/j.envsoft.2013.06.003>
- [92] Myerson, R.B.: Optimal Auction Design. *Mathematics of Operations Research* **6**(1), 58–73 (1981)
- [93] Myerson, R.B., Satterthwaite, M.A.: Efficient mechanisms for bilateral trading. *Journal of Economic Theory* **29**(2), 265–281 (1983). [https://doi.org/10.1016/0022-0531\(83\)90048-0](https://doi.org/10.1016/0022-0531(83)90048-0)

- [94] Needham, J., Edkins, G., Pimpale, G., Bartsch, H., Hobbhahn, M.: Large language models often know when they are being evaluated (2025), <https://arxiv.org/abs/2505.23836>
- [95] Ng Yi Sheng, E., Shen, B.: Formal mechanisms for market stability in self-interested agent societies: A marketplace simulation study. arXiv preprint arXiv:2607.08652 (2026), <https://arxiv.org/abs/2607.08652>
- [96] Nickerson, R.C., Varshney, U., Muntermann, J.: A method for taxonomy development and its application in information systems. *European Journal of Information Systems* **22**(3), 336–359 (2013). <https://doi.org/10.1057/ejis.2012.26>
- [97] Niranjani, H., Kumar, U., Tan, P.X.: Internal vs. external: Comparing deliberation and evolution for multi-agent constitutional design (2026), <https://arxiv.org/abs/2605.09128>
- [98] North, D.C.: *Institutions, Institutional Change and Economic Performance. Political Economy of Institutions and Decisions*, Cambridge University Press, Cambridge (1990). <https://doi.org/10.1017/CB09780511808678>
- [99] Nowak, M.A.: Five rules for the evolution of cooperation. *Science* **314**(5805), 1560–1563 (2006). <https://doi.org/10.1126/science.1133755>, <https://www.science.org/doi/abs/10.1126/science.1133755>
- [100] Ostrom, E.: *Governing the Commons: The Evolution of Institutions for Collective Action. Political Economy of Institutions and Decisions*, Cambridge University Press, Cambridge (1990). <https://doi.org/10.1017/CB09780511807763>
- [101] Ostrom, E.: *Understanding Institutional Diversity*. Princeton University Press, Princeton, NJ (2005). <https://doi.org/10.1515/9781400831739>
- [102] Ostrom, E.: Background on the institutional analysis and development framework. *Policy Studies Journal* **39**(1), 7–27 (2011). <https://doi.org/10.1111/j.1541-0072.2010.00394.x>
- [103] Ostrom, E., Walker, J., Gardner, R.: Covenants with and without a sword: Self-governance is possible. *American Political Science Review* **86**(2), 404–417 (1992). <https://doi.org/10.2307/1964229>
- [104] Papadimitriou, C.: Algorithms, games, and the internet. In: *Proceedings of the Thirty-Third Annual ACM Symposium on Theory of Computing*. pp. 749–753. STOC '01, Association for Computing Machinery, New York, NY, USA (Jul 2001). <https://doi.org/10.1145/380752.380883>
- [105] Piatti, G., Jin, Z., Kleiman-Weiner, M., Schölkopf, B., Sachan, M., Mihalcea, R.: Cooperate or collapse: Emergence of sustainable cooperation in a society of LLM agents. In: *Proceedings of the 38th International Conference on Neural Information Processing Systems. NIPS '24*, Curran Associates Inc., Red Hook, NY, USA (2024)
- [106] Popper, N.: Lost Passwords Lock Millionaires Out of Their Bitcoin Fortunes - The New York Times (Mar 2021), <https://web.archive.org/web/20210321104514/https://www.nytimes.com/2021/01/12/technology/bitcoin-passwords-wallets-fortunes.html>, accessed 30th of June 2026
- [107] Premack, D., Woodruff, G.: Does the chimpanzee have a theory of mind? *Behavioral and Brain Sciences* **1**(4), 515–526 (Dec 1978). <https://doi.org/10.1017/S0140525X00076512>

- [108] Rehm, G.B.: From certain doom to survival: Agent-driven self-governance in LLM agent societies (2026), <https://arxiv.org/abs/2609.22600>
- [109] Resnick, P., Kuwabara, K., Zeckhauser, R., Friedman, E.: Reputation systems. *Commun. ACM* **43**(12), 45–48 (Dec 2000). <https://doi.org/10.1145/355112.355122>
- [110] Schelling, T.C.: *The Strategy of Conflict*. Harvard University Press, Cambridge, MA (1960)
- [111] Seyedin, F., Weller, A., Yun, J., Babaei, M.: The politician, the liar, and the obedient worker: Emerging behavior of LLM agents in hierarchical games. *arXiv preprint arXiv:2608.09574* (2026), <https://arxiv.org/abs/2608.09574>
- [112] Shapira, N., Levy, M., Alavi, S.H., Zhou, X., Choi, Y., Goldberg, Y., Sap, M., Shwartz, V.: Clever Hans or neural theory of mind? Stress testing social reasoning in large language models. In: Graham, Y., Purver, M. (eds.) *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*. pp. 2257–2273. Association for Computational Linguistics, St. Julian’s, Malta (Mar 2024). <https://doi.org/10.18653/v1/2024.eacl-long.138>, <https://aclanthology.org/2024.eacl-long.138/>
- [113] Shoham, Y., Tennenholtz, M.: On social laws for artificial agent societies: Off-line design. *Artificial Intelligence* **73**(1–2), 231–252 (1995)
- [114] Skalse, J., Howe, N.H.R., Krashennnikov, D., Krueger, D.: Defining and characterizing reward hacking. In: *Advances in Neural Information Processing Systems*. vol. 35, pp. 9460–9471. Curran Associates, Inc. (2022), https://proceedings.neurips.cc/paper_files/paper/2022/hash/3d719fee332caa23d5038b8a90e81796-Abstract-Conference.html
- [115] Sofroniew, N., Kauvar, I., Saunders, W., Chen, R., Henighan, T., Hydrie, S., Citro, C., Pearce, A., Tarn, J., Gurnee, W., Batson, J., Zimmerman, S., Rivoire, K., Fish, K., Olah, C., Lindsey, J.: Emotion concepts and their function in a large language model (2026), <https://arxiv.org/abs/2604.07729>
- [116] Sorensen, T., Moore, J., Fisher, J., Gordon, M.L., Miresghallah, N., Rytting, C.M., Ye, A., Jiang, L., Lu, X., Dziri, N., Althoff, T., Choi, Y.: Position: A roadmap to pluralistic alignment. In: Salakhutdinov, R., Kolter, Z., Heller, K.A., Weller, A., Oliver, N., Scarlett, J., Berkenkamp, F. (eds.) *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21–27, 2024*. pp. 46280–46302. *Proceedings of Machine Learning Research, PMLR / OpenReview.net* (2024), <https://proceedings.mlr.press/v235/sorensen24a.html>
- [117] Steinberger, P.: OpenClaw: Your own personal AI assistant. <https://github.com/openclaw/openclaw> (2026), accessed: 2026-06-24
- [118] Syrnikov, M.B., Pierucci, F., Galisai, M., Prandi, M., Bisconti, P., Giarrusso, F., Sorokoletova, O., Suriani, V., Nardi, D.: Institutional AI: Governing LLM collusion in multi-agent Cournot markets via public governance graphs (2026), <https://arxiv.org/abs/2601.11369>

- [119] Tang, X., Kim, K., Song, Y., Lothritz, C., Li, B., Ezzini, S., Tian, H., Klein, J., Bissyandé, T.F.: CodeAgent: Autonomous Communicative Agents for Code Review. In: Al-Onaizan, Y., Bansal, M., Chen, Y.N. (eds.) *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. pp. 11279–11313. Association for Computational Linguistics, Miami, Florida, USA (Nov 2024). <https://doi.org/10.18653/v1/2024.emnlp-main.632>
- [120] Tewolde, E., Zhang, X., Guzman Piedrahita, D., Conitzer, V., Jin, Z.: CoopEval: Benchmarking cooperation-sustaining mechanisms and LLM agents in social dilemmas. In: *Proceedings of the 43rd International Conference on Machine Learning*. *Proceedings of Machine Learning Research*, vol. 306, pp. 120222–120287. PMLR (2026), <https://proceedings.mlr.press/v306/tewolde26a.html>
- [121] The Robot Report Staff: Serve robotics has deployed 2,000+ delivery robots across the U.S. The Robot Report (Dec 2025), <https://www.therobotreport.com/serve-robotics-has-deployed-2000-delivery-robots-a-cross-u-s/>, accessed: 2026-05-07
- [122] Tomasello, M.: *A Natural History of Human Morality*. Harvard University Press (2016)
- [123] Tomuschat, C.: The Legacy of Nuremberg. *Journal of International Criminal Justice* 4(4), 830–844 (Sep 2006). <https://doi.org/10.1093/jicj/mql051>
- [124] Trivedi, R.S., Jaques, N., Cross, L., Vezhnevets, A.S., Leibo, J.Z.: Position: Solipsistic superintelligence is unlikely to be cooperative. In: *Proceedings of the 43rd International Conference on Machine Learning*. *Proceedings of Machine Learning Research*, vol. 306, pp. 172306–172329. PMLR (2026), <https://proceedings.mlr.press/v306/trivedi26a.html>
- [125] Truong, V.Q., Ritchie, M.D.: Eras of bioinformatics technologies from command-line interfaces to artificial intelligence (AI) chatbots. *Briefings in Bioinformatics* 27(4), bbag256 (2026). <https://doi.org/10.1093/bib/bbag256>
- [126] United States Court of Appeals for the Second Circuit: In re: Citibank August 11, 2020 wire transfers (Sep 2022), <https://www.govinfo.gov/content/pkg/USCOURTS-ca2-21-00487/pdf/USCOURTS-ca2-21-00487-0.pdf>, docket No. 21-487, decided September 8, 2022
- [127] U.S. Securities and Exchange Commission: SEC approves new exchange rules for breaking clearly erroneous trades (Oct 2009), <https://www.sec.gov/newsroom/press-releases/2009-215-sec-approves-new-exchange-rules-breaking-clearly-erroneous-trades>
- [128] Vallinder, A., Hughes, E.: Cultural evolution of cooperation among LLM agents. arXiv preprint arXiv:2412.10270 (2024), <https://arxiv.org/abs/2412.10270>
- [129] Vedanta S P, Kumaraguru, P.: I can’t believe it’s corrupt: Evaluating corruption in multi-agent governance systems. arXiv preprint arXiv:2603.18894 (2026), <https://arxiv.org/abs/2603.18894>

- [130] Vezhnevets, A.S., Agapiou, J.P., Aharon, A., Ziv, R., Matyas, J., Duéñez-Guzmán, E.A., Cunningham, W.A., Osindero, S., Karmon, D., Leibo, J.Z.: Generative agent-based modeling with actions grounded in physical, social, or digital space using Concordia (2023), <https://arxiv.org/abs/2312.03664>
- [131] Walker, A.: Citibank sues for \$176m payment made in error. BBC News (Aug 2020), <https://www.bbc.co.uk/news/business-53802994>, accessed: 2026-06-09
- [132] Wang, A., Morgenstern, J., Dickerson, J.P.: Large language models that replace human participants can harmfully misportray and flatten identity groups. *Nat. Mac. Intell.* **7**(3), 400–411 (2025). <https://doi.org/10.1038/S42256-025-00986-Z>, <https://doi.org/10.1038/s42256-025-00986-z>
- [133] Wang, Y., Chen, Y., Meng, F., Chen, X., Yang, X., Zhang, M.: Law in silico: Simulating legal society with LLM-based agents. In: Findings of the Association for Computational Linguistics: ACL 2026. pp. 8061–8104. Association for Computational Linguistics, San Diego, California, United States (2026). <https://doi.org/10.18653/v1/2026.findings-acl.396>, <https://aclanthology.org/2026.findings-acl.396/>
- [134] Weidinger, L., Rauh, M., Marchal, N., Manzini, A., Hendricks, L.A., Mateos-Garcia, J., Bergman, S., Kay, J., Griffin, C., Bariach, B., Gabriel, I., Rieser, V., Isaac, W.: Sociotechnical safety evaluation of generative AI systems. arXiv preprint arXiv:2310.11986 (2023), <https://arxiv.org/abs/2310.11986>
- [135] van der Weij, T., Hofstätter, F., Jaffe, O., Brown, S.F., Ward, F.R.: AI sandbagging: Language models can strategically underperform on evaluations. In: The Thirteenth International Conference on Learning Representations (ICLR) (2025), <https://openreview.net/forum?id=7Qa2SpjxIS>
- [136] Willis, R., Zhao, J., Du, Y., Leibo, J.Z.: Evaluating collective behaviour of hundreds of LLM agents (2026), <https://arxiv.org/abs/2602.16662>
- [137] Wooldridge, M., Jennings, N.R.: Intelligent agents: Theory and practice. *The Knowledge Engineering Review* **10**(2), 115–152 (1995). <https://doi.org/10.1017/S0269888900008122>
- [138] Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., Jiang, L., Zhang, X., Zhang, S., Liu, J., Awadallah, A.H., White, R.W., Burger, D., Wang, C.: AutoGen: Enabling next-gen LLM applications via multi-agent conversations. In: First Conference on Language Modeling (COLM) (2024), <https://openreview.net/forum?id=BAakY1hNKS>
- [139] Wu, Y., Liang, S., Zhang, C., Wang, Y., Zhang, Y., Guo, H., Tang, R., Liu, Y.: From Human Memory to AI Memory: A Survey on Memory Mechanisms in the Era of LLMs (Apr 2025). <https://doi.org/10.48550/arXiv.2504.15965>
- [140] Yalim, J.: Toward Dynamically Controlling Slurm’s Classic Fairshare Algorithm. In: Practice and Experience in Advanced Research Computing 2020: Catch the Wave. pp. 538–542. PEARC ’20, Association

- for Computing Machinery, New York, NY, USA (Jul 2020). <https://doi.org/10.1145/3311790.3399628>
- [141] Yang, M., Fan, S., Yu, K., Wang, D., Sun, L., Peng, H., Yu, P.S., Liu, Z.: Paying for honesty without knowing the truth: Reputation-penalty design for LLM marketplace agents (2026), <https://arxiv.org/abs/2607.28330>
 - [142] Zhou, J., Lu, T., Mishra, S., Brahma, S., Basu, S., Luan, Y., Zhou, D., Hou, L.: Instruction-following evaluation for large language models (2023)
 - [143] Zhu, K., Du, H., Hong, Z., Yang, X., Guo, S., Wang, Z., Wang, Z., Qian, C., Tang, X., Ji, H., You, J.: MultiAgentBench: Evaluating the collaboration and competition of LLM agents. In: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). pp. 8580–8622. Association for Computational Linguistics, Vienna, Austria (2025). <https://doi.org/10.18653/v1/2025.acl-long.421>, <https://aclanthology.org/2025.acl-long.421/>
 - [144] Zhu, S., Lin, Y., Kaistha, S., Li, W., Wang, B., Zha, H., Hadfield, G.K., Poupart, P.: Talk, judge, cooperate: Gossip-driven indirect reciprocity in self-interested LLM agents (2026), <https://arxiv.org/abs/2602.07777>